

PayFix AI 2.0 Business, User and Solution Requirements

Field	Value
Document	Business, User and Solution Requirements
Project	PayFix AI 2.0
Version	1.0
Status	Requirements – In Analysis
Owner	Roshan Thakur
Requirement Basis	Discovery + To-Be Process Design
Strategic Direction	OPT-06 Hybrid
Core Foundation	OPT-02 Process / Rules / Workflow
AI Position	Conditional and evidence-gated
Scenario	Synthetic Portfolio Project
Business Approval	Not claimed
Requirement Validation	Progressive / portfolio-based

1. Purpose

This document defines and manages the PayFix AI 2.0 business, user and solution requirements derived from the Discovery evidence and approved portfolio-level To-Be process design.

The purpose is to convert identified business problems, user needs, process improvements, control boundaries and data requirements into clear, testable and traceable requirements.

Requirements will be defined before detailed implementation or technology selection wherever practical, while recognising that some requirements legitimately reflect the selected To-Be operating model.

AI is not treated as a default requirement. AI-specific requirements will only be created for use cases retained through the earlier AI suitability assessment.

1.1 Requirements Management Objectives

ID	Objective
RMO-01	Maintain traceability from Discovery evidence to requirements
RMO-02	Separate business outcomes from solution behaviour
RMO-03	Preserve user needs without unnecessarily prescribing technology
RMO-04	Make requirements clear and testable
RMO-05	Identify priority explicitly
RMO-06	Identify assumptions and dependencies
RMO-07	Maintain human/control boundaries

RMO-08	Include accessibility and assisted-service needs
RMO-09	Include data, audit and measurement requirements
RMO-10	Prevent unsupported AI requirements
RMO-11	Support later acceptance criteria and testing
RMO-12	Support Jira backlog refinement and delivery planning

2. Requirements Hierarchy

Business Problems / Opportunities



Business Requirements



User Requirements



Solution Requirements

- ├— Functional Requirements
- ├— Data Requirements
- ├— UX / Accessibility Requirements
- ├— Control Requirements
- └— Non-Functional Requirements



Conditional AI Requirements



Acceptance Criteria / Test Evidence

Requirements are related hierarchically but not necessarily one-to-one. One business requirement may support several user requirements, and one user requirement may require multiple functional, data, accessibility or control requirements.

3. Requirement Types

Prefix	Requirement Type	Purpose
BR	Business Requirement	Defines the business outcome or capability required
UR	User Requirement	Defines what a user needs to achieve
FR	Functional Requirement	Defines behaviour the future service/system must provide
DR	Data Requirement	Defines information, data, provenance or event needs
UXR	UX / Accessibility Requirement	Defines usability, communication and inclusive-service needs
CR	Control Requirement	Defines authority, audit, oversight, exception or control needs
NFR	Non-Functional Requirement	Defines quality attributes such as security, availability and performance
AIR	AI-Specific Requirement	Defines constraints/behaviour applicable only to an approved AI use case

Conventional automation requirements are normally captured as FRs supported by relevant CRs/NFRs rather than creating a separate “automation requirement” category.

3.1 Business Requirements — BR

Business requirements describe what PayFix AI 2.0 must enable at an outcome or business-capability level.

They should not normally specify screens, APIs, databases, models or detailed implementation methods.

Example only:

BR-001 — The future dispute service must provide clear ownership of active dispute cases throughout the lifecycle.

Not:

“The system must use MongoDB to save owners.”

3.2 User Requirements — UR

User requirements translate the provisional user needs established during UX Discovery into requirements describing what customers or operational users need to accomplish.

They remain centred on the user's goal rather than the implementation.

Example:

UR-CUS-001 — A customer needs to understand whether any action is required from them while their dispute is open.

3.3 Functional Requirements — FR

Functional requirements define what the future service or supporting digital capability must do to enable the business and user requirements.

Example:

FR-001 — The service shall record the current queue responsible for each active dispute case.

3.4 Data Requirements — DR

Data requirements define what information must be captured, maintained, exchanged or traceable to support the process, controls and measurement.

Example:

DR-001 — Each ownership transfer shall record the previous queue, new queue, transfer timestamp and handoff reason.

3.5 UX and Accessibility Requirements — UXR

UX and accessibility requirements define service characteristics necessary for understandable interaction, assisted journeys, communication clarity and inclusive use.

Example:

UXR-001 — The service shall clearly indicate when customer action is required and what action must be completed.

3.6 Control Requirements — CR

Control requirements define required authority, oversight, audit, exception, segregation-of-duty and decision-boundary behaviour.

Example:

CR-001 — A required refund approval shall remain attributable to an authorised human approval role.

3.7 Non-Functional Requirements — NFR

Use categories such as:

- Security
- Privacy
- Availability
- Reliability

- Recoverability
- Performance
- Auditability
- Maintainability
- Accessibility
- Scalability where justified
- Observability

Numerical thresholds will only be specified where an evidence basis exists or where they are later established as an explicit design/test assumption.

3.8 AI-Specific Requirements — AIR

AIR requirements apply only to AI capabilities that survive the options and suitability assessment.

They define additional controls such as source traceability, human review, rejection/correction, fallback, evaluation logging and limits on AI decision authority.

AIR requirements do not imply that an AI use case has been approved for production deployment.

Example:

AIR-001 — An AI-generated evidence summary shall not replace the authoritative source evidence used to create the summary.

4. Requirement Identification Convention

Use:

Business

BR-001

User

For customers:

UR-CUS-001

For Customer Service:

UR-CS-001

For Payment Operations:

UR-OPS-001

For Fraud:

UR-FRA-001

For Finance:

UR-FIN-001

For Technical:

UR-TEC-001

For Complaints:

UR-COM-001

For Service / Process Owner:

UR-SVC-001

Solution

FR-001

DR-001

UXR-001

CR-001

NFR-001

AIR-001

IDs remain stable once assigned. If a requirement is withdrawn, its ID should not be reused for a different requirement.

5. Requirement Statement Standard

For solution requirements, use where practical:

The service shall [required behaviour].

For business requirements:

The future dispute service must enable [business outcome/capability].

For user requirements:

[User] needs to [goal] so that [reason/outcome].

Example:

UR-OPS-001 — A Payment Operations analyst needs to see the current case owner, lifecycle status and outstanding customer action so that they can understand the case's current position without reconstructing the history.

6. Requirement Quality Criteria

Each requirement should be:

RQ-01 — Necessary

It supports an evidenced need, process or control.

RQ-02 — Clear

A reader can understand what is required.

RQ-03 — Atomic

Avoid combining unrelated requirements.

RQ-04 — Testable

There is a way to determine whether it has been satisfied.

RQ-05 — Traceable

The source can be identified.

RQ-06 — Feasible

No known reason makes it impossible.

RQ-07 — Consistent

It does not contradict other requirements.

RQ-08 — Appropriate in detail

It is neither so vague that it is meaningless nor prematurely technical.

RQ-09 — Prioritised

Importance is recorded.

RQ-10 — Controlled

Status, assumptions and changes are visible.

6.1 Requirement Quality Examples

Poor

The system should be fast and user friendly.

Problems:

- subjective;
- not testable;
- combines concepts;
- no defined outcome.

Better

The service shall clearly display whether customer action is currently required on an active dispute.

Poor

AI should automatically handle fraud cases.

Problems:

- solution assumed;
- authority unclear;
- conflicts with control analysis;
- untestable scope.

Better

AI assistance shall not independently determine the authoritative Fraud outcome.

Poor

All cases should be automated.

Better

Stable, repeatable administrative activities may be automated where deterministic logic and safe exception handling are defined.

7. Requirement Prioritisation

Priority	Meaning
Must	Essential for the defined service/control or viable release
Should	Important but a viable service could operate temporarily without it
Could	Valuable but lower priority
Won't for current scope	Explicitly excluded/deferred from the current scope

“Must” should not become the default for every requirement.

7.1 Priority Decision Factors

Consider:

- business problem severity;
- user need;
- control necessity;
- dependency;
- accessibility;
- process criticality;
- data requirement;
- regulatory/control importance;
- MVP necessity;
- delivery feasibility;

- evidence confidence;
- alternative/manual capability.

8. Requirement Status

Status	Meaning
Draft	Initial requirement captured
Proposed	Analysed and suitable for validation
Validated — Portfolio	Consistent with available portfolio evidence/design
Deferred	Valid need but postponed
Rejected	Requirement considered and intentionally not retained
Superseded	Replaced by another requirement

Important:

“Approved by stakeholders” will not be used unless genuine stakeholder approval exists.

For this synthetic project, most finished requirements should eventually become:

Validated — Portfolio

not:

Business Approved

9. Evidence Confidence

Confidence	Meaning
High	Strong direct evidence
Medium	Supported by multiple relevant project artefacts

Low-Medium	Plausible but requires further validation
Low	Limited evidence / significant assumption

Requirement priority and evidence confidence are separate concepts. A requirement may be a Must because of a control boundary while still require real organisational validation.

10. Master Requirements Register Structure

Column	Purpose
Requirement ID	Stable identifier
Requirement Type	BR / UR / FR / DR / UXR / CR / NFR / AIR
Requirement	Formal statement
Source	Discovery / user need / To-Be / control etc.
Rationale	Why needed
Priority	Must / Should / Could / Won't
Status	Draft / Proposed / Validated etc.
Evidence Confidence	High / Medium etc.
Parent Requirement	Hierarchical linkage
Process Stage	TB-01–TB-14
User / Role	Relevant user
Related Control	TCP / CR where applicable
Jira Evidence	Future delivery item
Acceptance/Test Link	Later test/AC reference

Assumption / Dependency	Constraint
Notes	Additional context

11. Requirement Source Register

Source Type	Example
BASE	Synthetic baseline
STK	Stakeholder analysis
RC	Root-cause hypothesis
ASIS	As-Is process
PP	Pain point
IO	Improvement opportunity
UN	User need
UXF	UX finding
ASN	Accessibility/support need
OPT	Options appraisal
GATE	Mandatory gate
AIU	AI suitability assessment
TB	To-Be activity
GW	To-Be gateway
TCP	To-Be control point
PI	Process improvement
BIH	Benefit hypothesis
ENT	Data entity
IF-TB	Information flow
LAC	Architecture constraint

12. Requirements Traceability Model

Discovery Evidence



Business Requirement



User Requirement



Functional / Data / UX / Control / NFR



Jira Story / Task



Acceptance Criteria



Test Case / UAT



Implementation Evidence



KPI / Benefit Evidence

Traceability should work in both directions: from a requirement back to its business/user justification and from a source need forward to the features and tests intended to satisfy it.

12.1 Requirements Traceability Matrix — RTM

Column	Example
Source	UN-07
Business Requirement	BR-004
User Requirement	UR-OPS-002
Solution Requirement	FR-014
Process Stage	TB-05
Control	—
Jira	Future story

Acceptance Criteria	AC reference
Test Case	Future
Status	Draft / Validated
Evidence	Confluence/Jira

Sourc e	Busin ess Requir ement	User Requir ement	Soluti on Requir ement	Proce ss	Contr ol	Jira	AC/Te st	Status
PP-05 / UN- 01	BR- 002	Later 24C	Later	TB- 02/03	—	Later	Later	Valida ted — Portfol io
PP-02 / UN- 07	BR- 004	Later 24C	Later	TB-05	—	Later	Later	Valida ted — Portfol io
UN- 09 / UN-10	BR- 006	Later 24C	Later	TB- 08/09	TCP- 03/04	Later	Later	Valida ted — Portfol io
GATE- 01-03	BR- 007	Later 24C	Later	TB- 09/10/ 12	TCP- 04/06 /07/0 8	Later	Later	Valida ted — Portfol io
UN- 03 / UN- 04	BR- 008	Later 24C	Later	TB-11	—	Later	Later	Valida ted — Portfol io

UN-05 / UN-06	BR-009	Later 24C	Later	TB-01	Assisted path	Later	Later	Validated — Portfolio
UN-14 / UN-19	BR-010	Later 24C	Later	TB-14	Traceability	Later	Later	Validated — Portfolio
Step 22E	BR-016	Later 24C	Later	Selected stages	AI controls	Later	Later	Validated — Portfolio

Source	BR	UR	Solution Req	Process	Control	Jira	AC/Test	Status
UN-01	BR-002/005	UR-CUS-001	Later	TB-02/03/07	—	Later	Later	Validated — Portfolio
UN-03	BR-008	UR-CUS-003	Later	TB-05/11	—	Later	Later	Validated — Portfolio
UN-05	BR-009	UR-CUS-005	Later	TB-01	Accessibility	Later	Later	Validated — Portfolio
UN-07	BR-004/008/011	UR-CS-001	Later	TB-05/11	—	Later	Later	Validated — Portfolio

UN-08	BR-002/005/011	UR-OPS-001	Later	TB-06	—	Later	Later	Validated — Portfolio
UN-10	BR-006/007/011	UR-FRA-001	Later	TB-08/09F	TCP-03/04	Later	Later	Validated — Portfolio
UN-11	BR-006/007/011	UR-FIN-001	Later	TB-09R/10	TCP-05/06	Later	Later	Validated — Portfolio
UN-13	BR-011/013	UR-COM-001	Later	TB-12	TCP-08	Later	Later	Validated — Portfolio
UN-14	BR-010/011/017	UR-SVC-001	Later	TB-14	Traceability	Later	Later	Validated — Portfolio
UN-19	BR-010/017	UR-SVC-002	Later	TB-14	—	Later	Later	Validated — Portfolio

Sourc e	BR	UR	Soluti on Requir ement	Proce ss	Contr ol	Jira	AC/Te st	Status
UN-02	BR-008	UR-CUS-002	FR-006	TB-01	—	Later	Later	Validated — Portfolio

UN-04	BR-005/008	UR-CUS-004	FR-010/011/040	TB-05/07/11	—	Later	Later	Validated — Portfolio
UN-07	BR-004	UR-CS-001	FR-018/019	TB-05	—	Later	Later	Validated — Portfolio
UN-08	BR-002/005	UR-OPS-001	FR-020-025	TB-06/07	—	Later	Later	Validated — Portfolio
UN-10	BR-006/007	UR-FRA-001	FR-028/030/036	TB-08/09	TCP-03/04	Later	Later	Validated — Portfolio
UN-11	BR-006/007	UR-FIN-001	FR-032/037/038	TB-09/10	TCP-05/06	Later	Later	Validated — Portfolio
UN-13	BR-013	UR-COM-001	FR-044-046	TB-12	TCP-08	Later	Later	Validated — Portfolio
Step 22	BR-016	—	FR-026/050	Select ed	AI controls	Later	Later	Validated — Portfolio
UN-19	BR-017	UR-SVC-002	FR-051/052	TB-14	—	Later	Later	Validated — Portfolio

Source	BR	UR	FR	DR	Process	Status
UN-03	BR-008	UR-CUS-003	FR-039/041	DR-002/035	TB-11	Validated — Portfolio
UN-04	BR-005/008	UR-CUS-004	FR-010/011	DR-016/017	TB-07	Validated — Portfolio
UN-07	BR-004	UR-CS-001	FR-018/019	DR-018-021	TB-05	Validated — Portfolio
UN-08	BR-002/005	UR-OPS-001	FR-020-025	DR-008/011-015/023	TB-06/07	Validated — Portfolio
UN-10	BR-006/007	UR-FRA-001	FR-028/030/036	DR-026-030/032	TB-08/09	Validated — Portfolio
UN-11	BR-006/007	UR-FIN-001	FR-032/037	DR-026-028/034	TB-09/10	Validated — Portfolio
UN-14	BR-010/011	UR-SVC-001	FR-051	DR-018-025/039-044	TB-14	Validated — Portfolio
AIU-02	BR-016	—	FR-050	DR-048/049	TB-06	Conditional

Source	BR	UR	FR	DR	UXR	Processes	Status
--------	----	----	----	----	-----	-----------	--------

UN-01	BR-002/005	UR-CUS-001	FR-007/011	DR-014/015	UXR-001/002/012/013	TB-02/03/07	Validated — Portfolio
UN-03	BR-008	UR-CUS-003	FR-039/041	DR-002/035	UXR-007/010/011	TB-11	Validated — Portfolio
UN-04	BR-005/008	UR-CUS-004	FR-010/040	DR-016/017	UXR-008/009	TB-07/11	Validated — Portfolio
UN-05	BR-009	UR-CUS-005	FR-001/002	DR-010	UXR-016-022	TB-01	Validated — Portfolio
UN-07	BR-004/008	UR-CS-001	FR-018/039	DR-018/035	UXR-029/032	TB-05/11	Validated — Portfolio
UN-18	BR-004/006	UR-OPS-003	FR-019/029	DR-020/021	UXR-030/031	TB-08	Validated — Portfolio
AIU-02	BR-016	—	FR-050	DR-049	UXR-035/036	TB-06	Conditional

Source	BR	UR	FR	DR/UXR	CR/NFR/AIR	Processes	Status
--------	----	----	----	--------	------------	-----------	--------

GATE-01	BR-007	UR-FRA-001	FR-036	DR-032	CR-002 / AIR-009	TB-09F	Validated — Portfolio
GATE-02	BR-007	UR-FIN-001	FR-037	DR-034	CR-003/009/010 / AIR-010	TB-09R/10	Validated — Portfolio
TCP-08	BR-013	UR-COM-001	FR-046	DR-036	CR-004 / AIR-011	TB-12	Validated — Portfolio
GW-19	BR-012	—	FR-048	DR-043	CR-006/012	Exceptions	Validated — Portfolio
GW-16	BR-012/015	—	FR-049	DR-042	CR-014 / NFR-008	Fallback	Validated — Portfolio
UN-05	BR-009	UR-CUS-005	FR-001/002	UXR-016-022	CR-016	TB-01	Validated — Portfolio
Step 23F	BR-011	UR-SVC-001	FR-051	DR-039-044	NFR-015/016	TB-14	Validated — Portfolio

AIU-02	BR-016	—	FR-050	DR-049 / UXR-035/036	AIR-001-008/013-016	TB-06	Condi tional
--------	--------	---	--------	----------------------	----------------------------	-------	-----------------

13. Requirement Relationship Types

Relationship	Meaning
Derived from	Requirement comes from evidence/need
Supports	Requirement contributes to another requirement
Depends on	Cannot operate without another capability
Constrains	Places a limit/control on another requirement
Conflicts with	Potential contradiction requiring resolution
Supersedes	Replaces an earlier requirement
Tested by	Links to acceptance/test evidence
Delivered by	Links to Jira/delivery work

14. Requirement Dependencies

Examples:

Example 1

Deterministic routing depends on:

- reliable classification;
- route rules;
- structured data;
- queue definition.

Example 2

AIU-02 Evidence Summary depends on:

- evidence provenance;
- authorised source access;
- human reviewer;
- fallback;
- evaluation mechanism.

Example 3

Meaningful status depends on:

- reliable workflow state;
- customer-action state;
- ownership/event updates.

Dependencies should be made explicit rather than hidden inside the requirement statement.

15. Requirement Conflict Management

Potential conflicts may include:

- customer simplicity vs mandatory information;
- automation vs exception flexibility;
- transparency vs restricted Fraud information;
- faster processing vs required control;
- detailed audit vs user/admin burden;
- AI assistance vs privacy/data minimisation;
- standardisation vs accessible/assisted exceptions.

Resolution approach:

1. Identify conflict.
2. Link affected requirements.
3. Identify stakeholders/perspectives.
4. Assess business/user/control impact.
5. Consider alternatives.
6. Record decision and rationale.

7. Update affected requirements.

16. Requirement Assumptions

ID	Assumption
RA-01	Future workflow can conceptually maintain case state
RA-02	Explicit ownership can be represented
RA-03	Relevant business rules can be defined later
RA-04	Specialist roles remain available
RA-05	Appropriate source evidence can be accessed by authorised users
RA-06	Assisted-service route remains available
RA-07	Reason/event information can be captured
RA-08	Automation can fall back to manual processing
RA-09	AI remains optional
RA-10	Real technical architecture requires later validation
RA-11	Real organisational approval has not occurred
RA-12	Requirements may change following testing/prototype evidence

17. Requirement Constraints

ID	Constraint
----	------------

RCN-01	Fraud authority must remain appropriately controlled
RCN-02	Required refund approval authority must be retained
RCN-03	Complaint outcomes remain appropriately human-owned
RCN-04	Sensitive decisions require accountable human ownership
RCN-05	Assisted/human service must remain available
RCN-06	AI cannot be required for the core service
RCN-07	Generated AI content is not authoritative source evidence
RCN-08	Technology failure requires safe fallback
RCN-09	Important process decisions must be appropriately traceable
RCN-10	Access to information must be controlled
RCN-11	No specific vendor/model/platform is mandated yet
RCN-12	Requirements must remain testable

18. Requirement Validation Approach

For every requirement ask:

Business validity

Does it solve an evidenced problem or enable required capability?

User validity

Does it support a documented user need?

Process validity

Does it fit the To-Be flow?

Control validity

Does it preserve required authority and safeguards?

Data validity

Is the information available or is a new data requirement needed?

Technical feasibility

Is there an obvious feasibility issue?

Testability

Could later acceptance criteria prove it?

Scope validity

Does it belong in PayFix AI 2.0?

18.1 Verification vs Validation

Concept	Question
Requirements Verification	Is the requirement written correctly?
Requirements Validation	Is this the right requirement for the business/user need?

Verification checks:

- clear;
- atomic;
- consistent;
- testable;
- correctly identified.

Validation checks:

- solves relevant need;

- reflects process;
- acceptable to stakeholders;
- preserves controls;
- appropriate priority.

18.2 Portfolio Validation Limitation

This project does not claim direct approval from real PayFix executives, operational teams or customers.

Requirements can be checked for consistency against the synthetic project evidence, user needs, process design and control model and labelled **Validated — Portfolio** where appropriate.

In a live organisation, formal validation and approval would require the relevant business, operational, customer, control and technology stakeholders.

19. Requirement Change Control

Change identified



Requirement affected?



Record proposed change



Assess source / process / user / control impact



Assess related requirements



Assess Jira / acceptance / testing impact



Accept / Modify / Reject



Update requirement version/status



Update traceability

20. Requirements Baseline Principle

A requirements baseline represents an agreed working set used for a delivery stage. It does not mean the requirements can never change.

For this portfolio, a baseline means the requirements have completed internal quality and traceability checks and are suitable to drive subsequent backlog, acceptance and testing work. It does not represent genuine approval by a real organisation.

21. Requirement Decomposition Principle

Example:

Business requirement

BR-X

Clear ownership is required.

↓

User requirement

Operations user needs to see who currently owns the case.

↓

Functional requirement

Service records current owner.

↓

Data requirement

Owner ID/role + timestamp + history.

↓

Acceptance criteria

Given an ownership transfer, previous and new ownership can be demonstrated.

Decomposition should preserve the original business/user intent rather than create functionality without traceable purpose.

22. Requirement → Acceptance Criteria Principle

Formal acceptance criteria will be created during backlog/testing preparation rather than embedded prematurely into every high-level business requirement.

Functional and control requirements should ultimately have sufficient acceptance evidence to determine whether the requirement has been satisfied.

Given / When / Then

Example:

Given an active case is transferred to Fraud Review,
when the transfer completes,
then the new queue, handoff reason, transfer timestamp and return owner are recorded.

23. Requirements → Jira Delivery Model

BR / UR

↓

Feature / Epic where appropriate

↓

FR / DR / UXR / CR / NFR

↓

User Story / Task

↓

Acceptance Criteria

↓

Test / UAT

Not every requirement should automatically become a separate Jira ticket. Several closely related requirements may be delivered through one coherent story, while complex requirements may require several delivery items.

24. Requirements in Agile Delivery

Agile delivery does not remove the need for requirements analysis.

Requirements may progressively evolve from business/user needs into refined backlog items and acceptance criteria as delivery knowledge improves.

The BA should maintain traceability while avoiding excessive upfront detail for capabilities that are not yet near delivery.

24.1 Kanban Use

Potential Kanban work includes:

- requirement discovery;
- clarification;
- dependency investigation;
- defects;
- policy questions;
- data-quality issues;
- AI evaluation questions;
- post-release improvement.

Planned product increments can move into Scrum delivery once requirements are sufficiently understood, while unresolved analysis work may continue through Kanban.

25. Requirements Integrity Rules

Do not invent:

- specific regulatory obligations not established;
- real organisational approval limits;
- refund monetary thresholds;
- actual SLA targets not defined;
- production system names;
- specific vendors;
- AI model;
- cloud provider;
- arbitrary AI accuracy thresholds;
- fake customer preferences;
- fake stakeholder approvals;

- unsupported volume/cost benefits.

Where information is unavailable:

state assumption → create validation requirement/question → do not fabricate.

26. Requirements Currently Out of Scope

At this stage, do not define:

- detailed physical database schema;
- API payload specification;
- production infrastructure;
- cloud topology;
- AI model hyperparameters;
- final UI visual design;
- exact commercial costs;
- real legal interpretation;
- vendor-specific configuration.

27. Requirements Coverage Map

Evidence Area	Requirement Types Expected
Baseline problems	BR
UN-01–UN-19	UR / FR / UXR
To-Be activities TB-01–TB-14	FR
GW-01–GW-20	FR / CR
TCP-01–TCP-12	CR
ENT-01–ENT-20	DR
IF-TB01–20	DR / FR
Accessibility analysis	UXR / NFR
Automation fallback	FR / CR / NFR
AIU-01–06	AIR / FR / CR
Event model	DR / NFR

KPI framework	DR / FR
Technology failure	NFR / CR
Privacy/security	NFR / CR

27.1 Coverage Safeguard

Requirement completeness will be assessed against the Discovery and To-Be evidence rather than by simply counting the number of requirements.

A smaller set of clear, traceable and testable requirements is more valuable than a large catalogue of duplicated or speculative requirements.

28. Requirement Review Checklist

For each requirement confirm:

- unique ID;
- correct requirement type;
- clear statement;
- single primary purpose;
- traceable source;
- rationale;
- priority;
- status;
- evidence confidence;
- user/process linkage;
- control linkage where relevant;
- dependency recorded;
- no unnecessary technical assumption;
- testability;
- no contradiction;
- no fake approval;
- AI used only where justified.

29. Conclusion

The PayFix AI 2.0 requirements-management approach has been established before detailed requirements are written.

Requirements will be organised across business, user, functional, data, UX/accessibility, control, non-functional and conditional AI categories using stable identifiers and documented quality standards.

MoSCoW priority, evidence confidence, requirement status, assumptions, dependencies and traceability will be maintained separately to avoid confusing importance with certainty or approval.

Requirements will trace from Discovery and user needs through the To-Be process into Jira delivery items, acceptance criteria, testing and eventual benefit evidence.

AI-specific requirements will only be created for previously justified use cases and will retain human-review, source-traceability and fallback boundaries.

The framework is now ready for the first formal requirement set: PayFix AI 2.0 Business Requirements.

30. Business Requirements

Business requirements define the business capabilities and outcomes that the PayFix AI 2.0 future dispute service must enable.

They are derived from the synthetic baseline, stakeholder analysis, As-Is process, root-cause hypotheses, user needs, options appraisal and approved To-Be process design.

These requirements intentionally avoid prescribing specific vendors, applications, APIs, databases, AI models or implementation technologies.

The requirements are currently assessed against the portfolio evidence and do not represent approval from a real organisation.

30.1 Business Requirement Rule

A business requirement should describe **what business capability or outcome is required**, not exactly how a software solution must implement it.

Detailed behaviour will later be decomposed into Functional, Data, UX/Accessibility, Control, Non-Functional and conditional AI requirements.

31. BR-001 — End-to-End Dispute Lifecycle

Field	Value
Requirement ID	BR-001
Type	Business Requirement
Requirement	The future dispute service must enable an end-to-end payment-dispute lifecycle from initiation through investigation, specialist handling where required, decision, customer communication and closure.
Source	ASIS, TB-01-TB-13
Rationale	The service requires a controlled lifecycle rather than disconnected operational activities.
Priority	Must
Status	Validated — Portfolio
Evidence Confidence	Medium
Process Scope	TB-01-TB-13

Monitoring and continuous improvement are addressed separately under BR-016.

32. BR-002 — Appropriate Information Completeness

Field	Value
Requirement ID	BR-002

Requirement	The future dispute service must enable the appropriate information and evidence required for a case to be captured and assessed at the earliest suitable stage without unnecessarily preventing valid cases from progressing.
Source	PP-05, IO-02, UN-01, UN-08, TB-02, TB-03
Rationale	Discovery identified information completeness as a plausible contributor to rework and downstream effort.
Priority	Must
Status	Validated — Portfolio
Confidence	Medium

every possible piece of evidence must be collected at intake.

33. BR-003 — Consistent Classification and Routing

Field	Value
Requirement ID	BR-003
Requirement	The future dispute service must enable consistent classification and routing using defined business logic where suitable while supporting authorised handling of ambiguous and exceptional cases.
Source	PP-02, IO-01, UN-09, GW-04–GW-06, PI-03, PI-04

Rationale	Routing ambiguity can create unnecessary transfer and ownership uncertainty.
Priority	Must
Status	Validated — Portfolio
Confidence	Medium

34. BR-004 — Explicit Case Ownership

Field	Value
Requirement ID	BR-004
Requirement	Every active dispute case must have an identifiable current responsibility state, including an appropriate queue, role or owner responsible for its current progression.
Source	PP-02, PP-11, UN-07, UN-18, TB-05, PI-05
Rationale	Discovery identified ownership and handoff visibility as important process concerns.
Priority	Must
Status	Validated — Portfolio
Confidence	Medium

35. BR-005 — Controlled Information and Evidence Requests

Field	Value
Requirement ID	BR-005

Requirement	The future dispute service must enable additional customer information or evidence to be requested, tracked and returned to the appropriate case activity using an explicit reason and customer-action state.
Source	UN-01, UN-04, UN-15, TB-07, GW-07, LP-01, LP-02, PI-12
Rationale	Evidence requests should not create uncontrolled or repetitive loops.
Priority	Must
Status	Validated — Portfolio
Confidence	Medium

36. BR-006 — Structured Specialist Referral

Field	Value
Requirement ID	BR-006
Requirement	The future dispute service must enable specialist referrals to contain sufficient reason, requested action and relevant case context, with clear specialist ownership and an appropriate return path.
Source	PP-01, PP-02, PP-10, UN-09, UN-10, UN-18, TB-08, TB-09, PI-10, PI-11

Rationale	Specialist pathways, particularly Fraud Review, represented significant process friction in the baseline.
Priority	Must
Status	Validated — Portfolio
Confidence	Medium

37. BR-007 — Sensitive Human Decision Authority

Field	Value
Requirement ID	BR-007
Requirement	The future dispute service must retain appropriate accountable human authority for sensitive dispute decisions, Fraud outcomes, required refund approvals, complaint outcomes and relevant service exceptions.
Source	GATE-01–GATE-03, TCP-04, TCP-06–TCP-09, RDP-01, Step 22E, Step 23D
Rationale	Process improvement and automation must not remove required human accountability.
Priority	Must
Status	Validated — Portfolio
Confidence	Medium

“All decisions must be manual.”

38. BR-008 — Meaningful Customer Status and Action Information

Field	Value
Requirement ID	BR-008
Requirement	The future dispute service must enable customers to receive meaningful information about dispute progress, required customer action, relevant next steps and final outcome where appropriate.
Source	UN-02–UN-04, UN-16, UN-17, UXF-02, UXF-07, TB-11, PI-06, PI-15
Rationale	Customers should not need unnecessary contact simply to determine whether action is required.
Priority	Must
Status	Validated — Portfolio
Confidence	Medium

39. BR-009 — Assisted and Human Service Route

Field	Value
Requirement ID	BR-009
Requirement	The future dispute service must retain an appropriate assisted and human-supported route for customers for whom the standard journey is insufficient or unsuitable.
Source	UN-05, UN-06, ASN-01–ASN-10, GW-01, PI-21

Rationale	Accessibility and additional-support needs must not be displaced by automation.
Priority	Must
Status	Validated — Portfolio
Confidence	Medium

40. BR-010 — Measurable Waiting, Handoffs and Rework

Field	Value
Requirement ID	BR-010
Requirement	The future dispute service must enable material waiting, handoff and rework activity to be recorded with sufficient reason and lifecycle information to support operational management and process improvement.
Source	Step 20F, PP-02–PP-04, UN-14, UN-19, PI-07–PI-09, ENT-07–ENT-09
Rationale	The As-Is baseline could count several events but could not adequately explain why they occurred.
Priority	Must
Status	Validated — Portfolio
Confidence	Medium

41. BR-011 — End-to-End Decision and Audit Traceability

Field	Value
-------	-------

Requirement ID	BR-011
Requirement	The future dispute service must provide appropriate traceability of material case activities, ownership changes, specialist involvement, decisions, approvals, overrides and customer communications.
Source	UN-14, UN-19, TCP-01–TCP-12, ENT-05, ENT-12–ENT-19, PI-18
Rationale	Important business and customer-impacting actions should be explainable and reviewable.
Priority	Must
Status	Validated — Portfolio
Confidence	Medium

42. BR-012 — Safe Exception and Technology Fallback

Field	Value
Requirement ID	BR-012
Requirement	The future dispute service must enable valid cases to continue through an authorised exception or fallback route when standard rules, automation, system integration or optional AI assistance cannot safely complete an activity.
Source	GW-03, GW-16–GW-19, TEX-01–TEX-16, PI-19, LAC-06–LAC-07
Rationale	Technology or standardisation must not create silent process dead ends.

Priority	Must
Status	Validated — Portfolio
Confidence	Medium

43. BR-013 — Linked Complaint Handling

Field	Value
Requirement ID	BR-013
Requirement	The future dispute service must enable complaint and escalation handling to use the relevant dispute history, decision information and customer communication history without unnecessary reconstruction of the case.
Source	UN-13, TB-12, GW-14, PI-16
Rationale	Complaint handling should build on the existing case record rather than duplicate investigation effort unnecessarily.
Priority	Must
Status	Validated — Portfolio
Confidence	Medium

44. BR-014 — Closure Integrity

Field	Value
Requirement ID	BR-014

Requirement	The future dispute service must ensure that a dispute is only finalised when applicable decision, approval, customer communication and required business/control activities have been appropriately completed or accounted for.
Source	TB-13, GW-13, GW-15, TCP-10, PI-17
Rationale	Investigation completion should not be confused with complete end-to-end case closure.
Priority	Must
Status	Validated — Portfolio
Confidence	Medium

45. BR-015 — Selective Conventional Automation

Field	Value
Requirement ID	BR-015
Requirement	The future dispute service should enable stable and repeatable administrative activities to be automated where the underlying business logic, exception handling and recovery behaviour are sufficiently defined.
Source	OPT-03, Step 22F, TBP-05, PI-19

Rationale	Conventional automation may reduce administrative effort without requiring AI for deterministic activities.
Priority	Should
Status	Validated — Portfolio
Confidence	Medium

46. BR-016 — AI Independence and Evidence-Gated Use

Field	Value
Requirement ID	BR-016
Requirement	The core payment-dispute service must operate end-to-end without AI, and any AI-assisted capability must only be introduced where its incremental value, controls and human-review arrangements are sufficiently demonstrated.
Source	Step 22E, AIU-01–AIU-06, Step 22G, TBP-06, PI-20, LAC-07
Rationale	Options analysis did not demonstrate that AI is necessary for the core service.
Priority	Must
Status	Validated — Portfolio
Confidence	Medium

47. BR-017 — Performance Measurement and Continuous Improvement

Field	Value
-------	-------

Requirement ID	BR-017
Requirement	The future dispute service should enable reliable measurement of process performance and recurring friction so that service improvements can be identified, prioritised and evaluated against defined baseline evidence.
Source	UN-19, TB-14, PI-22, BIH-04, BIH-09, KPI-TB01-KPI-TB18
Rationale	PayFix should be capable of determining whether future changes actually improve the service.
Priority	Should
Status	Validated — Portfolio
Confidence	Medium

48. Business Requirements Register

ID	Business Requirement	Priority	Confidence	Status
BR-001	End-to-end dispute lifecycle	Must	Medium	Validated — Portfolio
BR-002	Appropriate information completeness	Must	Medium	Validated — Portfolio
BR-003	Consistent classification and routing	Must	Medium	Validated — Portfolio

BR-004	Explicit case ownership	Must	Medium	Validated — Portfolio
BR-005	Controlled information/evidence requests	Must	Medium	Validated — Portfolio
BR-006	Structured specialist referral	Must	Medium	Validated — Portfolio
BR-007	Sensitive human decision authority	Must	Medium	Validated — Portfolio
BR-008	Meaningful customer status/action information	Must	Medium	Validated — Portfolio
BR-009	Assisted/human service route	Must	Medium	Validated — Portfolio
BR-010	Measurable waiting/hand offs/rework	Must	Medium	Validated — Portfolio
BR-011	Decision/audit traceability	Must	Medium	Validated — Portfolio
BR-012	Safe exception and technology fallback	Must	Medium	Validated — Portfolio
BR-013	Linked complaint handling	Must	Medium	Validated — Portfolio

BR-014	Closure integrity	Must	Medium	Validated — Portfolio
BR-015	Selective conventional automation	Should	Medium	Validated — Portfolio
BR-016	AI-independent core + evidence-gated AI	Must	Medium	Validated — Portfolio
BR-017	Measurement and continuous improvement	Should	Medium	Validated — Portfolio

48.1 Priority Summary

Priority	Count
Must	15
Should	2
Could	0
Won't	0
Total	17

The relatively high number of Must business requirements reflects that this catalogue operates at business-capability level. Later functional requirements should use MoSCoW more selectively and should not inherit Must priority automatically.

49. Discovery Problem → Business Requirement Traceability

Discovery Area	Business Requirement
Information completeness	BR-002 / BR-005
Routing	BR-003
Ownership	BR-004
Waiting	BR-010 / BR-017
Handoffs	BR-004 / BR-006 / BR-010
Rework	BR-002 / BR-005 / BR-010
Fraud specialist friction	BR-006 / BR-007
Refund dependency	BR-007 / BR-011
Customer status/contact	BR-008
Additional support	BR-009
Complaint handling	BR-013
Closure	BR-014
Measurement	BR-010 / BR-017
Automation opportunities	BR-015
AI suitability	BR-016

50. Business Requirement → To-Be Process Traceability

BR	Primary To-Be Area
BR-001	TB-01–TB-13
BR-002	TB-02 / TB-03
BR-003	TB-04
BR-004	TB-05
BR-005	TB-07
BR-006	TB-08 / TB-09

BR-007	TB-09 / TB-10 / TB-12
BR-008	TB-01 / TB-07 / TB-11
BR-009	TB-01 / exception paths
BR-010	TB-05 / TB-07–TB-09 / TB-14
BR-011	TB-01–TB-14
BR-012	Exception/fallback paths
BR-013	TB-12
BR-014	TB-13
BR-015	Selected activities across TB-01–TB-14
BR-016	Optional AI assistance across selected stages
BR-017	TB-14

51. Business Requirement → User-Need Coverage

BR	Key User Needs
BR-002	UN-01, UN-08
BR-004	UN-07, UN-09, UN-18
BR-005	UN-01, UN-04, UN-15
BR-006	UN-09, UN-10, UN-11, UN-12, UN-18
BR-008	UN-02, UN-03, UN-04, UN-16, UN-17
BR-009	UN-05, UN-06
BR-010	UN-14, UN-19
BR-011	UN-13, UN-14, UN-19
BR-013	UN-13

BR-017	UN-19
--------	-------

Detailed one-to-one user-requirement mapping will be completed in Step 24C.

52. Business Requirements with Strong Control Implications

Business Requirement	Future Control Requirement Area
BR-003	Rule ownership, override
BR-004	Ownership traceability
BR-006	Specialist authority/transfer
BR-007	Human decision authority
BR-009	Human/assisted access
BR-011	Auditability
BR-012	Exception/fallback
BR-014	Closure controls
BR-015	Automation governance
BR-016	AI governance/human review

Detailed CR requirements will be created in Step 24G.

53. Business Requirements with Major Data Implications

BR	Main Data Need
BR-002	Intake/evidence completeness
BR-003	Classification/routing inputs
BR-004	Ownership history
BR-005	Customer-action/evidence-request data
BR-006	Referral/finding data

BR-008	Status/action/communication information
BR-010	Waiting/handoff/rework records
BR-011	Events/decisions/approvals/overrides
BR-013	Complaint/case linkage
BR-014	Closure state
BR-016	AI provenance/review
BR-017	KPI/event data

54. Business Requirement Dependencies

BR-003 Classification and Routing

Depends on:

- usable classification data;
- business rules;
- defined queues;
- exception process.

BR-004 Ownership

Depends on:

- lifecycle/workflow state;
- defined operational responsibility.

BR-006 Specialist Referral

Depends on:

- referral reasons;
- sufficient case context;
- specialist responsibilities;
- return ownership.

BR-008 Customer Status

Depends on:

- accurate workflow state;
- customer-action state;
- communication capability.

BR-010 Measurement

Depends on:

- structured event/reason data.

BR-015 Automation

Depends on:

- stable process;
- deterministic rules;
- data quality;
- exception/failure handling.

BR-016 AI

Depends on:

- approved use case;
- authoritative source information;
- human reviewer;
- fallback;
- evaluation evidence.

55. Business Requirement Tensions

Tension	Requirements
Complete information vs easy intake	BR-002 / BR-009
Standard routing vs exceptions	BR-003 / BR-012
Faster progression vs Fraud/control	BR-006 / BR-007
Automation vs human override	BR-015 / BR-012
Status transparency vs restricted information	BR-008 / BR-007 / BR-011

AI value vs data minimisation/control	BR-016 / BR-007 / BR-011
More measurement vs user burden	BR-010 / BR-017

These relationships are not necessarily contradictions. They identify areas where later detailed requirements must balance competing objectives.

56. Business Requirements Not Supported by Current Evidence

Proposed Statement	Position
“All dispute processing must use AI”	Rejected
“AI must automatically resolve disputes”	Rejected
“AI must approve refunds”	Rejected
“AI must decide Fraud outcomes”	Rejected
“All customer interaction must be self-service”	Rejected
“All handoffs must be removed”	Rejected
“All specialist waiting is waste”	Rejected
“All cases must be automated”	Rejected
“The service must use a specific vendor/platform”	Not supported
“The To-Be must achieve a specific percentage improvement immediately”	Not supported

57. Business Requirements Quality Review

Necessary

Each BR traces to Discovery, user need, process, control or option evidence.

Business-level

No BR specifies:

- API;
- database;
- cloud;
- model;
- UI component;
- specific technology.

Testable through decomposition

Although some BRs are high-level, each can later be decomposed into solution requirements and acceptance evidence.

Technology-neutral

AI and automation are only included where already justified by the options decision.

Human controls retained

BR-007 and BR-016 prevent inappropriate interpretation of the preferred hybrid option.

58. BA Interpretation

The Business Requirements translate the Discovery and future-state design into a stable set of business capabilities without prematurely prescribing detailed technical implementation.

The requirements demonstrate that the core PayFix AI 2.0 need is broader than AI: the service requires improved information quality, routing, ownership, evidence handling, specialist coordination, customer communication, traceability, accessibility and closure integrity.

Automation is positioned as a supporting capability for stable administrative work, while AI is constrained by a specific requirement that the core service remains operable without it.

The Business Requirements therefore provide a controlled foundation from which user and detailed solution requirements can now be derived.

59. Conclusion

Seventeen formal Business Requirements have been defined for PayFix AI 2.0.

Fifteen are currently prioritised as Must requirements because they represent core lifecycle, user, control, traceability or service-resilience capabilities, while selective conventional

automation and continuous improvement are currently classified as Should-level business capabilities.

Requirements have been traced to Discovery evidence, user needs, To-Be process stages and relevant control/data implications.

Unsupported requirements such as broad autonomous AI processing, AI-led Fraud decisions, AI refund approval, mandatory self-service and arbitrary performance targets have been explicitly excluded.

These Business Requirements now provide the parent capability layer for detailed User Requirements.

60. User Requirements

User Requirements define what customers and operational users need to achieve through the PayFix AI 2.0 future dispute service.

They formalise the provisional user needs identified during Step 21 User Research and UX Discovery and align them to the approved To-Be process and Business Requirements.

These requirements remain portfolio-validated rather than real-user validated. No customer interview, operational interview or stakeholder approval is claimed where none occurred.

User Requirements describe user goals and outcomes. Detailed solution behaviour will be defined later through Functional, Data, UX/Accessibility, Control and Non-Functional Requirements.

60.1 User Requirement Statement Standard

Use:

[User] needs to [achieve something] so that [user/business outcome].

Example:

A customer needs to understand whether any action is required from them so that they can progress the dispute without unnecessary uncertainty or contact.

Avoid:

“The customer needs a blue dashboard with a status widget.”

61. UR-CUS-001 — Understand Required Information and Evidence

Field	Value
Requirement ID	UR-CUS-001
User	Customer
Requirement	A customer needs to understand what information or evidence is required for their dispute so that they can provide the appropriate information without unnecessary repetition or uncertainty.
Source	UN-01 , UXH-02, UXF-03
Parent BR	BR-002 , BR-005
Process	TB-02, TB-03, TB-07
Priority	Must
Status	Validated — Portfolio
Evidence Confidence	Medium-Low
Rationale	Information completeness and evidence requests were identified as important journey issues, but direct real-user validation remains outstanding.

62. UR-CUS-002 — Submission Confirmation

Field	Value
Requirement ID	UR-CUS-002
User	Customer

Requirement	A customer needs confirmation that their dispute has been received and information about the immediate next step so that they know the dispute journey has started.
Source	UN-02 , CJ-02
Parent BR	BR-001, BR-008
Process	TB-01
Priority	Must
Status	Validated — Portfolio
Confidence	Medium-Low

63. UR-CUS-003 — Meaningful Case Status

Field	Value
Requirement ID	UR-CUS-003
Requirement	A customer needs meaningful information about the current position of their dispute so that they can understand its progress without relying on unnecessary support contact.
Source	UN-03 , UXF-02, CJ findings
Parent BR	BR-008
Process	TB-05, TB-07, TB-09, TB-11
Priority	Must
Status	Validated — Portfolio
Confidence	Medium-Low

“Meaningful status” does not require disclosure of restricted Fraud, security or internal-control information.

64. UR-CUS-004 — Know When Customer Action Is Required

Field	Value
Requirement ID	UR-CUS-004
Requirement	A customer needs to know whether they are required to take an action, what action is required and why where appropriate, so that avoidable delay or uncertainty is reduced.
Source	UN-04 , UXF-02
Parent BR	BR-005 , BR-008
Process	TB-05, TB-07, TB-11
Priority	Must
Status	Validated — Portfolio
Confidence	Medium-Low

65. UR-CUS-005 — Accessible and Assisted Journey

Field	Value
Requirement ID	UR-CUS-005
Requirement	A customer who requires additional support needs an appropriate accessible or assisted route so that the standard digital journey does not prevent them from progressing their dispute.
Source	UN-05 , ASN-01–ASN-10, UXF-08

Parent BR	BR-009
Process	TB-01 and exception routes
Priority	Must
Status	Validated — Portfolio
Confidence	Medium-Low

This requirement does not infer a disability, condition or vulnerability from customer data.

66. UR-CUS-006 — Human Assistance

Field	Value
Requirement ID	UR-CUS-006
Requirement	A customer needs access to appropriate human assistance when self-service, automation or another standard service route is insufficient for their situation.
Source	UN-06 , UXF-09, GW-01
Parent BR	BR-009, BR-012
Process	TB-01, exception/fallback pathways
Priority	Must
Status	Validated — Portfolio
Confidence	Medium-Low

67. UR-CUS-007 — Progress During Long-Running or Specialist Cases

Field	Value
Requirement ID	UR-CUS-007

Requirement	A customer with a long-running or specialist dispute needs appropriate progress information so that internal specialist processing does not leave them without meaningful understanding of the case position.
Source	UN-16, UXF-04
Parent BR	BR-006, BR-008
Process	TB-08, TB-09, TB-11
Priority	Should
Status	Validated — Portfolio
Confidence	Medium-Low

Progress information remains subject to appropriate Fraud, privacy and control restrictions.

68. UR-CUS-008 — Understand Final Outcome and Next Steps

Field	Value
Requirement ID	UR-CUS-008
Requirement	A customer needs to understand the final dispute outcome, relevant explanation and available next steps so that they can understand what has happened and what they can do next.
Source	UN-17, UXF-07
Parent BR	BR-008, BR-013, BR-014
Process	TB-10, TB-11, TB-12

Priority	Must
Status	Validated — Portfolio
Confidence	Medium-Low

69. UR-CUS-009 — Avoid Unnecessary Repetition

Field	Value
Requirement ID	UR-CUS-009
Requirement	A customer needs information and evidence already appropriately provided to be reused where suitable so that they are not unnecessarily required to repeat the same information during the dispute journey.
Source	UN-15 , UXF-03, UXF-06
Parent BR	BR-002, BR-005
Process	TB-02, TB-06, TB-07, TB-08
Priority	Should
Status	Validated — Portfolio
Confidence	Medium-Low

Reuse does not prevent a legitimate re-request where information is incomplete, outdated, unsuitable or different evidence is required.

70. Customer Service User Requirements

70.1 UR-CS-001 — Case Status, Ownership and Required Action Visibility

Field	Value
Requirement ID	UR-CS-001

User	Customer Service
Requirement	A Customer Service user needs to understand the current dispute status, responsible queue or owner and whether customer action is required so that they can provide accurate support without reconstructing the case journey.
Source	UN-07 , UXF-02, UXF-05
Parent BR	BR-004, BR-008, BR-011
Process	TB-05, TB-07, TB-11
Priority	Must
Status	Validated — Portfolio
Confidence	Medium-Low

71. Payment Operations User Requirements

71.1 UR-OPS-001 — Sufficient Case and Evidence Context

Field	Value
Requirement ID	UR-OPS-001
User	Payment Operations
Requirement	A Payment Operations analyst needs sufficiently complete case information and relevant evidence so that they can investigate the dispute without unnecessary reconstruction or avoidable information requests.
Source	UN-08 , UXF-03, UXF-06
Parent BR	BR-002, BR-005, BR-011

Process	TB-06
Priority	Must
Status	Validated — Portfolio
Confidence	Medium-Low

71.2 UR-OPS-002 — Clear Specialist Referral

Field	Value
Requirement ID	UR-OPS-002
Requirement	A Payment Operations analyst needs clear specialist-referral criteria, ownership and return expectations so that specialist involvement can be initiated and completed without unnecessary ambiguity or transfer.
Source	UN-09 , UXF-04, UXF-05
Parent BR	BR-003, BR-004, BR-006
Process	TB-08, TB-09
Priority	Must
Status	Validated — Portfolio
Confidence	Medium-Low

71.3 UR-OPS-003 — Sufficient Handoff Context

Field	Value
Requirement ID	UR-OPS-003

Requirement	An operational user receiving a transferred dispute needs sufficient context about the reason for transfer, previous activity and requested next action so that they can continue the case without unnecessary reconstruction or clarification.
Source	UN-18 , PP-10, UXF-05, UXF-06
Parent BR	BR-004, BR-006, BR-010
Process	TB-08, TB-09
Priority	Must
Status	Validated — Portfolio
Confidence	Medium-Low

Although the ID is under Payment Operations, the requirement also applies conceptually to receiving specialist users.

72. Fraud Review User Requirement

72.1 UR-FRA-001 — Structured Fraud Referral Context

Field	Value
Requirement ID	UR-FRA-001
User	Fraud Review
Requirement	A Fraud reviewer needs a structured referral containing the relevant reason, case context, available evidence and requested specialist action so that they can perform Fraud Review without unnecessary case reconstruction.

Source	UN-10 , UXF-04
Parent BR	BR-006, BR-007, BR-011
Process	TB-08, TB-09F
Control	TCP-03, TCP-04
Priority	Must
Status	Validated — Portfolio
Confidence	Medium-Low

The requirement improves referral quality; it does not alter human Fraud decision authority.

73. Refund / Finance User Requirement

73. Refund / Finance User Requirement

Field	Value
Requirement ID	UR-FIN-001
User	Refund / Finance
Requirement	An authorised Refund / Finance user needs the relevant refund basis, supporting information and required approval context so that an appropriate financial-control decision can be made and recorded.
Source	UN-11
Parent BR	BR-006, BR-007, BR-011
Process	TB-09R, TB-10
Control	TCP-05, TCP-06

Priority	Must
Status	Validated — Portfolio
Confidence	Medium-Low

74. Technical Support User Requirement

74.1 UR-TEC-001 — Technical Referral Context

Field	Value
Requirement ID	UR-TEC-001
User	Technical Support
Requirement	A Technical Support user needs clear technical referral context, the reason for investigation and relevant case information so that the technical issue can be investigated without unnecessary reconstruction.
Source	UN-12
Parent BR	BR-006, BR-011
Process	TB-08, TB-09T
Priority	Must
Status	Validated — Portfolio
Confidence	Medium-Low

75. Complaints User Requirement

75.1 UR-COM-001 — Complete Dispute and Communication History

Field	Value
Requirement ID	UR-COM-001

User	Complaints
Requirement	A Complaints user needs access to the relevant dispute history, decisions and customer communications so that the complaint can be reviewed without unnecessarily reconstructing the underlying case.
Source	UN-13 , UXF-07
Parent BR	BR-011, BR-013
Process	TB-12
Control	TCP-08
Priority	Must
Status	Validated — Portfolio
Confidence	Medium-Low

76. Service and Process Owner User Requirements

76.1 UR-SVC-001 — Operational Traceability

Field	Value
Requirement ID	UR-SVC-001
User	Service / Process Owner
Requirement	A Service or Process Owner needs material dispute activities, ownership changes, handoffs, waiting, rework, decisions and overrides to be traceable so that operational performance and control can be understood.

Source	UN-14 , UXF-10
Parent BR	BR-010, BR-011, BR-017
Process	TB-01–TB-14
Priority	Must
Status	Validated — Portfolio
Confidence	Medium-Low

76.2 UR-SVC-002 — Structured Process-Event Information

Field	Value
Requirement ID	UR-SVC-002
Requirement	A Service or Process Owner needs reliable structured process-event information so that recurring delay, handoff, rework and service-performance patterns can be measured and prioritised for improvement.
Source	UN-19 , UXF-11
Parent BR	BR-010, BR-017
Process	TB-14
Priority	Should
Status	Validated — Portfolio
Confidence	Medium-Low

77. User Requirements Register

ID	User	Requirement Theme	Priority	Confidence	Status
----	------	-------------------	----------	------------	--------

UR-CUS-001	Customer	Information/evidence requirements	Must	Medium-Low	Validated — Portfolio
UR-CUS-002	Customer	Submission confirmation	Must	Medium-Low	Validated — Portfolio
UR-CUS-003	Customer	Meaningful status	Must	Medium-Low	Validated — Portfolio
UR-CUS-004	Customer	Customer action	Must	Medium-Low	Validated — Portfolio
UR-CUS-005	Customer	Accessible/assisted journey	Must	Medium-Low	Validated — Portfolio
UR-CUS-006	Customer	Human assistance	Must	Medium-Low	Validated — Portfolio
UR-CUS-007	Customer	Specialist/long-running progress	Should	Medium-Low	Validated — Portfolio
UR-CUS-008	Customer	Final outcome/next steps	Must	Medium-Low	Validated — Portfolio
UR-CUS-009	Customer	Reduce unnecessary repetition	Should	Medium-Low	Validated — Portfolio

UR-CS-001	Customer Service	Status/owner/action visibility	Must	Medium-Low	Validated — Portfolio
UR-OPS-001	Payment Operations	Case/evidence context	Must	Medium-Low	Validated — Portfolio
UR-OPS-002	Payment Operations	Specialist referral	Must	Medium-Low	Validated — Portfolio
UR-OPS-003	Operational User	Handoff context	Must	Medium-Low	Validated — Portfolio
UR-FRA-001	Fraud	Fraud referral context	Must	Medium-Low	Validated — Portfolio
UR-FIN-001	Finance	Refund/appraisal context	Must	Medium-Low	Validated — Portfolio
UR-TEC-001	Technical	Technical referral context	Must	Medium-Low	Validated — Portfolio
UR-COM-001	Complaints	Case/communication history	Must	Medium-Low	Validated — Portfolio
UR-SVC-001	Process Owner	Operational traceability	Must	Medium-Low	Validated — Portfolio
UR-SVC-002	Process Owner	Process-event measurement	Should	Medium-Low	Validated — Portfolio

78. Original User Need → Formal User Requirement Traceability

Original Need	Formal Requirement
UN-01	UR-CUS-001
UN-02	UR-CUS-002
UN-03	UR-CUS-003
UN-04	UR-CUS-004
UN-05	UR-CUS-005
UN-06	UR-CUS-006
UN-07	UR-CS-001
UN-08	UR-OPS-001
UN-09	UR-OPS-002
UN-10	UR-FRA-001
UN-11	UR-FIN-001
UN-12	UR-TEC-001
UN-13	UR-COM-001
UN-14	UR-SVC-001
UN-15	UR-CUS-009
UN-16	UR-CUS-007
UN-17	UR-CUS-008
UN-18	UR-OPS-003
UN-19	UR-SVC-002

79. User Requirement → Business Requirement Traceability

User Requirement	Parent BR
UR-CUS-001	BR-002, BR-005

UR-CUS-002	BR-001, BR-008
UR-CUS-003	BR-008
UR-CUS-004	BR-005, BR-008
UR-CUS-005	BR-009
UR-CUS-006	BR-009, BR-012
UR-CUS-007	BR-006, BR-008
UR-CUS-008	BR-008, BR-013, BR-014
UR-CUS-009	BR-002, BR-005
UR-CS-001	BR-004, BR-008, BR-011
UR-OPS-001	BR-002, BR-005, BR-011
UR-OPS-002	BR-003, BR-004, BR-006
UR-OPS-003	BR-004, BR-006, BR-010
UR-FRA-001	BR-006, BR-007, BR-011
UR-FIN-001	BR-006, BR-007, BR-011
UR-TEC-001	BR-006, BR-011
UR-COM-001	BR-011, BR-013
UR-SVC-001	BR-010, BR-011, BR-017
UR-SVC-002	BR-010, BR-017

79.1 Business Requirement Coverage Observation

User Requirements do not need a one-to-one relationship with every Business Requirement.

Some Business Requirements, such as selective automation, AI independence and closure integrity, primarily constrain the future solution or operating model and will therefore receive stronger traceability through Functional, Control, Data, Non-Functional and AI requirements later in Step 24.

80. User Group Coverage

User Group	Number of URs
Customer	9
Customer Service	1
Payment Operations / receiving operational user	3
Fraud Review	1
Refund / Finance	1
Technical Support	1
Complaints	1
Service / Process Owner	2
Total	19

Requirement count does not indicate stakeholder importance. Different roles naturally require different levels of user-need decomposition.

81. Customer Requirements → Journey Stages

Customer Requirement	Main Journey Stage
UR-CUS-001	Initiation / evidence
UR-CUS-002	Submission
UR-CUS-003	Investigation / waiting / specialist
UR-CUS-004	Evidence/action stages
UR-CUS-005	Across journey
UR-CUS-006	Across journey
UR-CUS-007	Specialist / long-running case
UR-CUS-008	Outcome / complaint
UR-CUS-009	Evidence / investigation

82. User Requirement Tensions

Tension	Requirements / Controls
Clear status vs restricted Fraud information	UR-CUS-003 / BR-007
Easy intake vs information completeness	UR-CUS-001 / BR-002
Avoid repetition vs evidence may legitimately change	UR-CUS-009 / BR-005
Standardisation vs assisted support	UR-CUS-005 / BR-003
Automation efficiency vs human assistance	UR-CUS-006 / BR-015
Rich handoff context vs unnecessary data sharing	UR-OPS-003 / future privacy requirements
Complete case history vs role-based access	UR-COM-001 / future security requirements
Measurement vs operational data burden	UR-SVC-002 / future DR/NFR

83. User Requirement Design Implications

Customer

- clear information guidance;
- confirmation;
- status;
- customer-action visibility;
- assisted support;
- outcome explanation.

Operational users

- case context;
- ownership;

- structured handoff;
- source evidence;
- specialist context.

Specialist users

- relevant referral information;
- clear requested action;
- decision authority.

Process owners

- traceable events;
- waiting/handoff/rework information;
- measurable performance.

84. User Requirements Not Supported by Current Evidence

Proposed Requirement	Position
Customer wants an AI chatbot for every dispute	Not supported
Customer wants AI to decide the outcome	Not supported
Customer wants a specific mobile app	Not established
Operations wants all cases automated	Not supported
Fraud wants autonomous AI decisioning	Not supported
Customers prefer no human contact	Not established
All customers want the same service route	Not supported
Users require a specific vendor/system	Not established

User needs should not be invented from the technology concept.

85. User Requirement Evidence Confidence

Current User Requirements are assigned **Medium-Low evidence confidence** because they are strongly traceable to Discovery analysis, process evidence and role-based user hypotheses but have not been directly validated through real-user interviews or observational research.

This does not mean the requirements are low priority. Several remain Must requirements because the future design needs to support the underlying journey or control capability.

In a live project, interviews, usability research, operational walkthroughs and prototype testing would be used to increase or challenge the current confidence level.

86. User Requirements Requiring Future Validation

Validation Priority	Requirements
High	UR-CUS-001, 003, 004, 005, 006, 007, 008
High	UR-CS-001
High	UR-OPS-001, 002, 003
High	UR-FRA-001
High	UR-FIN-001
Medium-High	UR-TEC-001
High	UR-COM-001
Medium-High	UR-SVC-001, 002

“High validation priority” means evidence should be strengthened, not that the requirement is more important than all others.

86.1 Potential Validation Methods

User Group	Suitable Future Method
------------	------------------------

Customer	Interview / usability testing
Additional-support customer	Inclusive research / assisted-journey testing
Customer Service	Interview / workflow walkthrough
Payment Operations	Contextual interview / process walkthrough
Fraud	Specialist walkthrough
Finance	Control/process walkthrough
Technical	Referral walkthrough
Complaints	Case-history walkthrough
Process Owner	KPI/data review

87. User Requirement → Future Solution Requirement Examples

UR-CUS-004

May later derive:

- functional requirement for customer-action state;
- data requirement for action type;
- communication requirement;
- accessibility requirement.

UR-OPS-003

May derive:

- referral/handoff functionality;
- handoff-reason data;
- ownership history;
- optional AIU-03 requirements.

UR-SVC-002

May derive:

- event data requirements;
- reporting requirements;

- data-quality requirements;
- audit requirements.

Formal solution decomposition begins in Step 24D.

88. BA Interpretation

The User Requirements preserve the needs identified during UX Discovery while making the distinction between user goals and solution functionality explicit.

Customer requirements focus on understandable information, meaningful status, customer-action clarity, human/assisted support, reduced unnecessary repetition and understandable outcomes.

Operational and specialist requirements focus on sufficient case context, explicit ownership, structured transfers, relevant evidence and decision information.

Service-owner requirements focus on process traceability and measurable operational evidence.

AI is not represented as a user need because Discovery did not establish that users inherently require AI. AI may later be used only where it is an appropriate means of satisfying one or more underlying requirements.

89. Conclusion

Nineteen formal PayFix AI 2.0 User Requirements have been created, providing complete traceability to UN-01 through UN-19 from the earlier UX Discovery stage.

Requirements cover customers, Customer Service, Payment Operations, Fraud Review, Refund / Finance, Technical Support, Complaints and Service / Process Owner needs.

Each requirement has been linked to relevant Business Requirements and future-state process areas while avoiding premature specification of screens, technologies or AI solutions.

Requirements are labelled Validated — Portfolio with Medium-Low evidence confidence because they are supported by the synthetic portfolio evidence but have not been validated through real-user interviews.

The Business and User Requirement layers are now sufficiently established to begin defining detailed Functional Requirements.

90. Functional Requirements

Functional Requirements define the behaviour that the PayFix AI 2.0 future service must provide to satisfy the Business Requirements, User Requirements and approved To-Be process.

The requirements are derived from BR-001–BR-017, the nineteen formal User Requirements, TB-01–TB-14, GW-01–GW-20 and the future-state control and responsibility model.

Functional Requirements remain technology-neutral. They specify required service behaviour without selecting a case-management platform, workflow product, rules engine, automation platform, AI model, database or vendor.

Detailed data attributes, non-functional qualities, accessibility rules and AI-specific controls are defined separately in later requirement sections.

90.1 Functional Requirement Standard

Functional Requirements use the statement form:

The service shall [observable required behaviour].

Each requirement should describe a single primary behaviour wherever practical and should be capable of later decomposition into acceptance criteria and test evidence.

91. Intake and Case Creation Requirements

ID	Functional Requirement	Parent	Source	Priority
FR-001	The service shall enable a payment dispute to be initiated through an available standard or assisted service route.	BR-001, BR-009	UR-CUS-005/006, TB-01	Must

FR-002	The service shall enable an authorised Customer Service user to assist a customer with dispute initiation where assistance is required.	BR-009	UR-CUS-005/006, GW-01	Must
FR-003	The service shall create a unique dispute case when a valid initiation is recorded.	BR-001	TB-01	Must
FR-004	The service shall associate the dispute case with the relevant customer and transaction reference where available.	BR-001, BR-002	UR-CUS-001, TB-02	Must

FR-005	The service shall capture the customer's dispute description and relevant initiation information without replacing the customer's original information with generated content.	BR-002, BR-011	UR-CUS-001, TB-02	Must
FR-006	The service shall provide confirmation that the dispute has been received together with appropriate immediate next-step information.	BR-008	UR-CUS-002, TB-01	Must

FR-001 and FR-002 deliberately preserve both standard and assisted entry routes. The future service is therefore not dependent on a single digital self-service journey.

92. Completeness and Customer-Action Requirements

ID	Functional Requirement	Parent	Source	Priority
FR-007	The service shall assess whether the minimum information required for the current process stage is available.	BR-002	UR-CUS-001, TB-03, GW-02	Must
FR-008	Where required information is missing, the service shall identify that additional information or evidence is required rather than silently progressing the case as complete.	BR-002, BR-005	UR-CUS-001, TB-03	Must

FR-009	The service shall support an authorised human exception where a case is permitted to progress despite incomplete information.	BR-002, BR-012	GW-03, TEX-02	Must
FR-010	The service shall maintain whether customer action is currently required and the applicable type of customer action.	BR-005, BR-008	UR-CUS-004, TB-05/07	Must

FR-011	When additional customer information or evidence is required, the service shall issue an appropriate request and return the received information to the relevant case activity.	BR-005	UR-CUS-001/004, TB-07, LP-01/02	Must
---------------	---	--------	---------------------------------	-------------

Completeness is assessed relative to the current processing stage. FR-007 does not require every possible future piece of evidence to be collected during initial intake.

93. Classification, Routing and Ownership Requirements

ID	Functional Requirement	Parent	Source	Priority
FR-012	The service shall apply defined deterministic classification logic where the available information is sufficient for such logic.	BR-003	TB-04, GW-04	Must

FR-013	Where deterministic classification cannot appropriately determine a classification , the service shall route the case for authorised human review.	BR-003, BR-012	UR-OPS-002, GW-04/05	Must
FR-014	An authorised user shall be able to confirm or correct an ambiguous case classification .	BR-003	GW-05, OVR-02	Must
FR-015	The service shall apply defined priority logic where appropriate priority criteria are available.	BR-003	TB-04	Must

FR-016	The service shall apply defined routing logic where sufficient routing information and approved rules are available.	BR-003	TB-04, GW-06	Must
FR-017	Where standard routing cannot determine an appropriate destination, the service shall support authorised manual routing.	BR-003, BR-012	GW-06, TEX-05	Must
FR-018	The service shall assign an appropriate queue, role or owner to each active case requiring internal processing.	BR-004	UR-CS-001, TB-05	Must

FR-019	The service shall enable authorised ownership transfer while preserving the ability to identify the previous responsibility and reason for transfer.	BR-004, BR-010	UR-OPS-003, TB-05/08	Must
---------------	--	----------------	----------------------	-------------

Routing automation is subordinate to approved business logic. FR-016 does not authorise AI to become the default routing decision-maker.

94. Investigation and Evidence Requirements

ID	Functional Requirement	Parent	Source	Priority
FR-020	The service shall enable an authorised Payment Operations user to access the relevant case information required for investigation.	BR-001, BR-011	UR-OPS-001, TB-06	Must

FR-021	The service shall enable authorised users to access relevant available evidence associated with the dispute.	BR-002, BR-011	UR-OPS-001, TB-06	Must
FR-022	The service shall enable the investigator to determine whether available evidence is sufficient to progress toward an outcome.	BR-007	UR-OPS-001, GW-12	Must

FR-023	Where additional information or evidence is required during investigation, the service shall return the case to the appropriate evidence-request activity rather than restarting the entire dispute journey.	BR-005	UR-CUS-009, LP-02	Must
FR-024	The service shall support reuse of relevant information or evidence already appropriately provided where suitable.	BR-002, BR-005	UR-CUS-009	Should

FR-025	Where information or evidence must legitimately be requested again, the service shall support recording the reason for the repeat request.	BR-005, BR-010	UR-CUS-001/009	Should
FR-026	The service shall allow investigation activity to continue without requiring any optional AI assistance capability.	BR-016	Step 22, TB-06	Must

95. Specialist Referral and Review Requirements

ID	Functional Requirement	Parent	Source	Priority
----	------------------------	--------	--------	----------

FR-027	The service shall enable an investigator to initiate a specialist referral where specialist involvement is required.	BR-006	UR-OPS-002, TB-08, GW-08	Must
FR-028	Before a specialist referral is transferred, the service shall support validation that the required referral context is available.	BR-006	UR-FRA-001, UR-FIN-001, UR-TEC-001, TB-08	Must
FR-029	The service shall enable the referral to specify the specialist type, referral reason and requested specialist action.	BR-006, BR-011	UR-OPS-003, TB-08	Must

FR-030	The service shall support referral to Fraud Review when applicable Fraud referral criteria or authorised judgement indicate that Fraud Review is required.	BR-006, BR-007	UR-FRA-001, GW-09	Must
FR-031	The service shall support referral to Technical Support where technical investigation is required.	BR-006	UR-TEC-001, GW-10	Must
FR-032	The service shall support referral to Refund / Finance where a financial review or approval is required.	BR-006, BR-007	UR-FIN-001, GW-11	Must

FR-033	The service shall support more than one relevant specialist pathway where the case legitimately requires multiple specialist activities.	BR-006	GW-08-11, Step 23C	Should
---------------	--	--------	--------------------	---------------

95.1 Specialist Return Behaviour

Specialist activity must have a defined return path. The receiving specialist should not become an indefinite final holding queue simply because ownership was transferred.

This principle will later be supported by detailed data requirements for:

- specialist start;
- specialist completion;
- findings;
- return owner.

96. Decision and Approval Requirements

ID	Functional Requirement	Parent	Source	Priority
----	------------------------	--------	--------	----------

FR-034	The service shall enable authorised users to record specialist findings and make them available to the relevant case decision activity.	BR-006, BR-011	UR-FRA/FIN/TEC, TB-09	Must
FR-035	The service shall enable an accountable human user to record the authoritative dispute decision and supporting rationale where required.	BR-007, BR-011	TB-10, TCP-07	Must

FR-036	The service shall ensure that a Fraud outcome requiring specialist judgement is recorded under the authority of an appropriate human Fraud role.	BR-007	UR-FRA-001, TCP-04	Must
FR-037	The service shall ensure that a required refund approval is completed by an appropriately authorised human role before the dependent financial action is treated as approved.	BR-007	UR-FIN-001, TCP-05/06	Must

FR-038	The service shall identify whether required business or control approvals remain outstanding before progressing to final outcome completion.	BR-007, BR-014	GW-13	Must
---------------	--	----------------	-------	-------------

FR-035–FR-038 deliberately distinguish **decision support** from **decision authority**. Optional automation or AI may support information handling, but those capabilities do not become the accountable decision-maker.

97. Customer Communication Requirements

ID	Functional Requirement	Parent	Source	Priority
FR-039	The service shall enable appropriate customer communication to be triggered by relevant dispute lifecycle events.	BR-008	UR-CUS-002/003/004	Must

FR-040	The service shall enable the customer to be informed when a material action is required from them.	BR-005, BR-008	UR-CUS-004	Must
FR-041	The service shall support appropriate progress communication for long-running or specialist disputes where such communication is suitable.	BR-008	UR-CUS-003/007	Should

FR-042	The service shall enable the final dispute outcome, relevant explanation and appropriate next steps to be communicated to the customer.	BR-008, BR-014	UR-CUS-008	Must
FR-043	The service shall maintain a relationship between customer communications and the relevant dispute case.	BR-011, BR-013	UR-CS-001, UR-COM-001	Must

98. Complaint and Closure Requirements

ID	Functional Requirement	Parent	Source	Priority
----	------------------------	--------	--------	----------

FR-044	The service shall enable a complaint or escalation to be associated with the relevant underlying dispute case.	BR-013	UR-COM-001, TB-12	Must
FR-045	The service shall enable an authorised Complaints user to access the relevant dispute, decision and customer communication history required for complaint review.	BR-011, BR-013	UR-COM-001	Must

FR-046	The service shall enable the authoritative complaint outcome and rationale to be recorded by an appropriate human Complaints role.	BR-007, BR-013	TCP-08	Must
FR-047	Before final closure, the service shall validate that applicable decision, approval, communication and outstanding-action conditions have been appropriately completed or accounted for.	BR-014	TB-13, GW-15	Must

99. Exception, Fallback and Monitoring Requirements

ID	Functional Requirement	Parent	Source	Priority
----	------------------------	--------	--------	----------

FR-048	The service shall support authorised human override of applicable rule, routing, automation or AI-supported outcomes where an approved exception path exists.	BR-012	GW-19, OVR-01-10	Must
FR-049	Where a conventional automated activity fails, the service shall support a visible manual or recoverable fallback path so that the case is not silently blocked.	BR-012, BR-015	GW-16, TEX-11/12	Must

FR-050	Where optional AI assistance is unavailable, rejected or unsuitable, the corresponding business activity shall remain completable through the normal non-AI process.	BR-012, BR-016	GW-17/18, TEX-13/14	Must
FR-051	The service shall enable material lifecycle activity required for operational monitoring and process analysis to be captured or made available for reporting.	BR-010, BR-017	UR-SVC-001/002, TB-14	Must

FR-052	The service shall enable recurring operational friction or process-performance findings to be identified for subsequent service-improvement work.	BR-017	UR-SVC-002, TB-14, PI-22	Should
---------------	---	--------	--------------------------	---------------

100. Functional Requirements Register

Range	Requirement Area	Count
FR-001-006	Intake and Case Creation	6
FR-007-011	Completeness and Customer Action	5
FR-012-019	Classification, Routing and Ownership	8
FR-020-026	Investigation and Evidence	7
FR-027-033	Specialist Referral and Review	7
FR-034-038	Decision and Approval	5
FR-039-043	Customer Communication	5
FR-044-047	Complaints and Closure	4

FR-048-052	Exceptions, Fallback and Monitoring	5
Total		52

100.1 Functional Requirement Status

Requirements will move to **Validated — Portfolio** after the Step 24D quality, traceability and conflict checks have been completed.

101. Functional Requirement Priority Summary

Priority	Count
Must	44
Should	8
Could	0
Won't	0
Total	52

Functional Must priority is driven largely by the viable end-to-end lifecycle and existing control boundaries. Later backlog prioritisation may package or phase requirements differently; a Must requirement does not imply that each requirement becomes an independent Jira ticket.

102. Business Requirement → Functional Requirement Coverage

Business Requirement	Main Functional Coverage
BR-001 End-to-end lifecycle	FR-001-006, 020, 027, 034-047
BR-002 Completeness	FR-004/005, 007-009, 020-025
BR-003 Classification/routing	FR-012-017
BR-004 Ownership	FR-018-019
BR-005 Evidence requests	FR-008, 010-011, 023-025
BR-006 Specialist referral	FR-027-034

BR-007 Human authority	FR-009, 035–038, 046, 048
BR-008 Customer status/action	FR-006, 010–011, 039–043
BR-009 Assisted route	FR-001–002
BR-010 Waiting/handoff/rework	FR-019, 025, 029, 051
BR-011 Traceability	FR-005, 019–021, 029, 034–035, 043, 045
BR-012 Fallback	FR-009, 013, 017, 048–050
BR-013 Complaints	FR-043–046
BR-014 Closure	FR-038, 042, 047
BR-015 Automation	FR-016, 039, 049
BR-016 AI independence	FR-026, FR-050
BR-017 Continuous improvement	FR-051–052

Some Business Requirements are satisfied by combinations of functional, data, control, UX and non-functional requirements rather than Functional Requirements alone.

103. User Requirement → Functional Requirement Traceability

User Requirement	Main FR Coverage
UR-CUS-001	FR-005, 007–011, 023–025
UR-CUS-002	FR-003, FR-006
UR-CUS-003	FR-039, FR-041
UR-CUS-004	FR-010, FR-011, FR-040
UR-CUS-005	FR-001, FR-002
UR-CUS-006	FR-001, FR-002, FR-048–050
UR-CUS-007	FR-041
UR-CUS-008	FR-042, FR-044–047
UR-CUS-009	FR-023–025

UR-CS-001	FR-018-019, FR-039-043
UR-OPS-001	FR-020-026
UR-OPS-002	FR-012-019, FR-027-033
UR-OPS-003	FR-019, FR-028-029, FR-033-034
UR-FRA-001	FR-028-030, FR-034, FR-036
UR-FIN-001	FR-028-029, FR-032, FR-034, FR-037-038
UR-TEC-001	FR-028-029, FR-031, FR-034
UR-COM-001	FR-043-046
UR-SVC-001	FR-019, FR-034-035, FR-043, FR-048-051
UR-SVC-002	FR-051-052

104. To-Be Process Stage → Functional Requirement Coverage

Process Stage	Main FRs
TB-01 Initiation	FR-001-006
TB-02 Information Capture	FR-004-005
TB-03 Completeness	FR-007-011
TB-04 Classification/Routing	FR-012-017
TB-05 Ownership	FR-018-019
TB-06 Investigation	FR-020-026
TB-07 Additional Information	FR-010-011, FR-023-025
TB-08 Specialist Referral	FR-027-033
TB-09 Specialist Review	FR-030-034
TB-10 Decision/Approval	FR-034-038
TB-11 Communication	FR-039-043

TB-12 Complaints	FR-044-046
TB-13 Closure	FR-047
TB-14 Monitoring	FR-051-052

105. Gateway → Functional Requirement Traceability

Gateway	FR
GW-01 Assisted support?	FR-001-002
GW-02 Information complete?	FR-007-008
GW-03 Incomplete exception?	FR-009
GW-04 Deterministic classification?	FR-012-013
GW-05 Manual classification?	FR-014
GW-06 Routing determined?	FR-016-017
GW-07 More evidence?	FR-011, FR-023
GW-08 Specialist required?	FR-027
GW-09 Fraud required?	FR-030
GW-10 Technical required?	FR-031
GW-11 Finance required?	FR-032
GW-12 Evidence sufficient?	FR-022-023
GW-13 Controls complete?	FR-038
GW-14 Complaint?	FR-044
GW-15 Closure satisfied?	FR-047
GW-16 Automation successful?	FR-049
GW-17 AI applicable?	FR-050
GW-18 AI output acceptable?	FR-050 + future AIR
GW-19 Human override?	FR-048

106. Functional Requirements Requiring Detailed Data Support

- FR-003 case identifier;
- FR-004 customer/transaction linkage;
- FR-005 original dispute information;
- FR-007 completeness;
- FR-010 customer-action state;
- FR-019 ownership history;
- FR-021 evidence;
- FR-025 repeat-request reason;
- FR-029 specialist referral;
- FR-034 specialist findings;
- FR-035 decisions;
- FR-037 approval;
- FR-043 communications;
- FR-044 complaint linkage;
- FR-047 closure;
- FR-048 override;
- FR-051 process events.

These functional behaviours will be supported by formal DR requirements in Step 24E rather than duplicating detailed field specifications here.

107. Functional Requirements Requiring UX / Accessibility Support

Functional Area	Future UXR Need
FR-001/002	Assisted initiation
FR-006	Understandable confirmation
FR-008–011	Clear evidence/customer-action guidance
FR-018	Ownership/status visibility

FR-039-042	Understandable communication
FR-048-050	Usable exception/fallback
FR-041	Long-running case communication

108. Functional Requirements Requiring Control Requirements

FR	Future Control Area
FR-009	Incomplete-case exception
FR-012-017	Rule governance and override
FR-030	Fraud referral control
FR-035	Sensitive decision authority
FR-036	Fraud authority
FR-037	Refund authority
FR-038	Approval completeness
FR-046	Complaint authority
FR-047	Closure integrity
FR-048	Human override
FR-049	Automation failure
FR-050	AI fallback

109. Functional Requirements Potentially Supported by AI

Functional Area	Potential AI Use Case	AI Necessary?
Intake information	AIU-01	No
Evidence review	AIU-02	No
Specialist handoff	AIU-03	No
Investigation	AIU-04	No

Communication preparation	AIU-05	No
Ambiguous classification	AIU-06	No

No Functional Requirement is currently dependent on AI for successful core-service execution.

110. Conventional Automation Candidates

FR	Candidate Automation
FR-003	Case/reference creation
FR-006	Submission confirmation
FR-007	Completeness rule execution
FR-012	Standard classification
FR-015	Priority
FR-016	Standard routing
FR-018	Queue/work-item assignment
FR-039	Lifecycle communication triggers
FR-040	Customer-action notification
FR-047	Closure-condition validation
FR-051	Process-event capture

Candidate automation does not mean mandatory automation. Final automation scope depends on detailed rules, controls, feasibility and testing.

111. Human-Essential Functional Behaviour

Current design retains human authority for:

- FR-002 assisted customer support;
- FR-009 authorised incomplete-case exception;

- FR-014 ambiguous classification;
- FR-017 exceptional routing;
- FR-022 evidence sufficiency;
- FR-035 authoritative dispute decision;
- FR-036 Fraud judgement;
- FR-037 refund approval;
- FR-046 complaint decision;
- FR-048 authorised override.

Human-essential does not mean technology cannot provide supporting information or workflow assistance.

12. Functional Requirement Quality Review

Example check

FR-019

The service shall enable authorised ownership transfer while preserving the ability to identify the previous responsibility and reason for transfer.

This contains related behaviour belonging to one business event:

ownership transfer.

Therefore it remains acceptably atomic.

But avoid a requirement such as:

The service shall route cases, send emails, detect fraud, create refunds and generate reports.

112.1 Technology-Neutrality Check

Confirm no FR requires:

- Angular;
- React;
- MongoDB;
- SQL;
- AWS;

- Azure;
- OpenAI;
- a particular AI model;
- Power Automate;
- Jira;
- a named case-management vendor.

Technology choices may later be assessed against these requirements rather than becoming requirements simply because a product is available.

112.2 Functional vs Non-Functional Boundary

Examples:

Functional

The service shall enable a customer-action state to be maintained.

Non-functional

The service shall remain available according to an agreed availability requirement.

Functional

The service shall record a decision.

Non-functional

Decision records shall be protected from unauthorised access.

113. Functional Requirement Tensions

Tension	Requirements
Easy initiation vs completeness	FR-001-002 / FR-007-011
Deterministic routing vs exceptions	FR-012-017
Evidence reuse vs legitimate re-request	FR-024 / FR-025

Specialist efficiency vs required control	FR-027-038
Status transparency vs restricted information	FR-039-042
Automation vs manual fallback	FR-016 / FR-049
AI assistance vs AI independence	Future AI support / FR-026, FR-050
Closure automation vs exceptional cases	FR-047 / FR-048

These tensions will be addressed through Control, Data, UX and Non-Functional Requirements rather than by weakening one requirement informally.

114. Functional Requirements Not Supported

Proposed Function	Position
Automatically approve all disputes	Rejected
AI autonomously decides Fraud outcomes	Rejected
AI automatically approves refunds	Rejected
AI autonomously decides complaints	Rejected
Automatically close a case based only on AI output	Rejected
Remove all human support	Rejected
Route every case using AI	Not required
Require all evidence at initial intake	Rejected
Automatically remove all specialist referrals	Rejected
Automatically disclose unrestricted Fraud information to customers	Rejected

Require a specific technical platform	Not supported
---------------------------------------	----------------------

115. Example Functional Acceptance Decomposition

FR-018 — Assign ownership

Possible future acceptance:

Given a dispute is ready for internal processing,
when it enters the relevant queue,
then the current responsibility state is identifiable.

FR-029 — Specialist referral

Given specialist involvement is required,
when a referral is created,
then the specialist type, referral reason and requested action are available to the receiving role.

FR-050 — AI fallback

Given an optional AI capability is unavailable,
when the associated business activity is required,
then an authorised user can complete the activity through the normal non-AI process.

Formal acceptance criteria will be developed later against refined Jira stories and testing work.

116. BA Interpretation

The Functional Requirements translate business and user needs into observable future-service behaviour while maintaining separation from detailed implementation technology.

The catalogue covers the complete dispute lifecycle from initiation through information completeness, classification, routing, ownership, investigation, evidence handling, specialist processing, human decisions, customer communication, complaints, closure, exception handling and continuous improvement.

Deterministic rules and conventional automation are used where the underlying behaviour is stable and sufficiently defined, while authorised human pathways remain available for ambiguity, exceptions and sensitive decisions.

Importantly, no core Functional Requirement depends on AI. Potential AI assistance may later support selected functions, but FR-026 and FR-050 explicitly preserve a complete non-AI operating path.

117. Conclusion

Fifty-two Functional Requirements have been defined for PayFix AI 2.0 and traced to the Business Requirements, User Requirements, To-Be process stages and key decision gateways.

Functional coverage has been established across initiation, case creation, completeness, customer action, classification, routing, ownership, investigation, evidence, specialist referral, human decision-making, customer communication, complaint handling, closure, exception management, technology fallback and operational monitoring.

The requirements remain technology-neutral and do not prescribe a vendor, platform, database, cloud service or AI model.

Human accountability remains explicit for sensitive decisions and exceptions, while the core dispute lifecycle remains fully operable without AI.

Detailed information structures are intentionally deferred to the next requirements stage: Data and Information Requirements.

118. Data and Information Requirements

Data and Information Requirements define the information that PayFix AI 2.0 must capture, maintain, reference, exchange or make available to support the future payment-dispute process.

The requirements are derived from the Logical Business Data Entities ENT-01–ENT-20, information flows IF-TB01–IF-TB20, Functional Requirements FR-001–FR-052, process-event design and measurement framework.

These are logical information requirements rather than a physical database specification. They do not prescribe tables, collections, APIs, storage products, cloud platforms or database technologies.

Particular attention is given to source provenance, lifecycle history, ownership, waiting, handoff, rework, specialist activity, decisions, approvals, process events and the distinction between authoritative information and generated AI content.

118.1 Data Requirement Statement Standard

Data Requirements generally use forms such as:

The service shall capture...

The service shall maintain...

The service shall preserve...

The service shall make available...

The service shall distinguish...

Data Requirements specify the business information needed and its logical meaning without defining the physical implementation.

119. Case Identity and Lifecycle Data Requirements

ID	Data Requirement	Source	Related FR	Priority
DR-001	The service shall maintain a unique Case ID for each dispute case.	ENT-01	FR-003	Must
DR-002	The service shall maintain the current lifecycle status of each dispute case.	ENT-01, TB-01-13	FR-018, FR-047	Must
DR-003	The service shall record the date/time at which the dispute case is initiated.	ENT-01	FR-003	Must

DR-004	The service shall record the finalisation date/time where a case reaches an applicable final state.	ENT-01, ENT-20	FR-047	Must
DR-005	The service shall maintain the applicable case outcome and closure state where these are established.	ENT-01, ENT-20	FR-035, FR-047	Must
DR-006	The Case ID shall remain consistently referenceable across related case information, events, referrals, decisions, approvals, communications and other linked records.	LAC-01	Multiple FRs	Must

DR-006 describes logical referential consistency. It does not prescribe a relational database or specific technical key implementation.

119.1 Lifecycle-State Principle

The data model must distinguish an active or waiting case from a genuinely finalised case.

Pending is an active/open lifecycle condition and must not be interpreted as a completed case or as evidence that an SLA was met.

120. Customer, Transaction and Evidence Requirements

ID	Data Requirement	Source	Related FR	Priority
DR-007	The service shall maintain an appropriate customer reference linking the dispute to the relevant customer record where required.	ENT-01/02	FR-004	Must

DR-008	The service shall maintain the relevant transaction reference and authoritative transaction-information source where available.	ENT-03	FR-004, FR-020	Must
DR-009	The customer's original dispute description shall be preserved as authoritative customer-provided information.	ENT-01, TIP-13 principle	FR-005	Must
DR-010	The service shall record appropriate initiation-channel or assisted-service-route information where relevant to the journey.	ENT-02	FR-001/002	Must

DR-011	Each material Evidence Item shall be identifiable and associated with the relevant dispute case.	ENT-04	FR-021	Must
DR-012	Evidence records shall preserve appropriate provenance, including source and receipt information sufficient to understand where the evidence originated.	ENT-04, LAC-02	FR-021	Must

DR-013	Where evidence is replaced, corrected or superseded, the service should enable the relationship between the current and earlier evidence state to be understood where relevant.	ENT-04	FR-021/024	Should
---------------	---	--------	------------	---------------

120.1 Evidence Authority

Evidence summaries, notes or generated descriptions do not replace the authoritative underlying evidence.

Where summarisation is used, the relationship to the original source evidence must remain identifiable.

121. Completeness and Customer-Action Data Requirements

ID	Data Requirement	Source	Related FR	Priority
----	------------------	--------	------------	----------

DR-014	The service shall maintain an appropriate information-completeness state for the relevant stage of the case.	TB-03, GW-02	FR-007	Must
DR-015	Where information or evidence is missing, the service shall support recording the applicable missing-information or evidence reason.	TB-03/07	FR-008/011	Must
DR-016	The service shall distinguish whether customer action is currently required and the applicable action type.	UN-04, ENT data model	FR-010	Must

DR-017	The service should record the start and completion/end of material customer-action states where this information is required for journey or waiting analysis.	TB-07, KPI framework	FR-010/011	Should
---------------	---	----------------------	------------	---------------

122. Ownership, Waiting, Handoff and Rework Data Requirements

ID	Data Requirement	Source	Related FR	Priority
DR-018	The service shall maintain the current responsible queue, role or owner for each active internal case state where applicable.	ENT-06, TB-05	FR-018	Must

DR-019	The service shall preserve material ownership history, including ownership start and end information where applicable.	ENT-06	FR-019	Must
DR-020	A material ownership handoff shall record the previous responsibility , new responsibility , handoff time and handoff reason.	ENT-08	FR-019	Must

DR-021	A specialist or operational handoff shall support recording the requested action and intended return responsibility where relevant.	ENT-08/10	FR-029	Must
DR-022	A material waiting state shall support recording the waiting reason, start, end, dependency and accountable monitoring responsibility where applicable.	ENT-07	FR-051	Must

DR-023	Material rework shall support recording the rework reason, relevant activity and originating process stage.	ENT-09	FR-025, FR-051	Must
DR-024	The service should enable incorrect-routing corrections to be distinguished from legitimate planned handoffs.	HR-08, PI-04	FR-014/017/019	Should

DR-025	Waiting, handoff and rework reasons shall use sufficiently defined and governed values to support consistent operational interpretation and analysis.	WR/HR/RW frameworks	FR-051	Must
---------------	---	---------------------	--------	-------------

122.1 Observability Improvement

The As-Is data could identify that handoff, waiting or rework events occurred, but it had limited ability to explain the underlying reason or actual duration.

DR-020–DR-025 therefore improve **process observability** before any claim is made that the events themselves have reduced.

123. Specialist Referral and Finding Data Requirements

ID	Data Requirement	Source	Related FR	Priority
DR-026	Each material specialist referral shall be identifiable and linked to the applicable dispute case.	ENT-10	FR-027	Must

DR-027	The specialist referral shall identify the specialist type, referral reason and requested action.	ENT-10	FR-029-032	Must
DR-028	The referral shall reference the relevant authorised case context and evidence required for the specialist activity.	ENT-10	FR-028/029	Must
DR-029	The service shall record material specialist findings and applicable rationale or result information.	ENT-11	FR-034	Must

DR-030	Specialist-processing information shall enable the relevant referral start, completion and return responsibility to be understood.	ENT-10/11	FR-029/034	Must
DR-031	Where multiple specialist activities are required, the service should enable their relationships or required sequence to be understood where operationally relevant.	GW-08-11	FR-033	Should

123.1 Specialist Context Principle

Structured referral does not mean sending every available piece of case information to every specialist.

The receiving role should receive the information necessary for the authorised specialist task, subject to later privacy, security and access-control requirements.

124. Outcome and Customer-History Data Requirements

ID	Data Requirement	Source	Related FR	Priority
DR-032	A material dispute Decision Record shall identify the decision type, outcome, accountable decision role, decision time and applicable rationale.	ENT-12	FR-035/036	Must
DR-033	The service shall distinguish a proposed or supporting outcome from the authoritative final business decision where both exist.	ENT-12, LAC-02/03	FR-035	Must

DR-034	Required approval information shall identify the approval type, status, approving authority/role and applicable request/decision timestamps.	ENT-13	FR-037/038	Must
DR-035	Material customer communications shall be associated with the case and record appropriate communication type, trigger/channel and timestamp information.	ENT-14	FR-039-043	Must

DR-036	A complaint or escalation record shall retain its relationship to the relevant dispute case and applicable complaint outcome/rationale.	ENT-15	FR-044-046	Must
DR-037	Closure information shall identify the final state, closure reason and closure timestamp where applicable.	ENT-20	FR-047	Must
DR-038	The lifecycle data shall distinguish active states such as Pending/Waiting from final states such as Resolved, Rejected or Cancelled.	TB-13, baseline methodology	FR-047	Must

124.1 Decision Data Boundary

An AI recommendation, automated rule result or analyst draft must not be stored or presented in a way that makes it indistinguishable from an authoritative human decision where human authority is required.

125. Process Event and Execution Data Requirements

ID	Data Requirement	Source	Related FR	Priority
DR-039	Material lifecycle events shall be identifiable by Event ID, Case ID, event type and timestamp.	ENT-05	FR-051	Must
DR-040	Where relevant, process events shall preserve material status and ownership transitions rather than only the latest state.	ENT-05/06	FR-019/051	Must

DR-041	Material deterministic rule execution shall be traceable to the applicable rule identity/version and result where required for control, explanation or testing.	ENT-16	FR-012/015/016/038	Must
DR-042	Material automation execution shall support recording the execution result and applicable failure, retry or reconciliation status.	ENT-17	FR-049	Must

DR-043	An authorised override shall support recording the original result/state, revised result/state, override reason, authorised actor and timestamp where applicable.	ENT-19	FR-048	Must
DR-044	Process-event information shall enable the actor type to be distinguished where relevant, including Customer, Human User, Rule, Workflow, Automation and AI Assistance.	ENT-05	FR-051	Must

126. Measurement, Provenance and Data-Quality Requirements

ID	Data Requirement	Source	Related FR	Priority
DR-045	The service should make available sufficient lifecycle data to calculate defined service and process measures using clearly established denominators.	KPI-TB01-18	FR-051/052	Should
DR-046	The service should make available sufficiently structured reason information to analyse waiting, handoff, rework and relevant customer-contact patterns.	BIH-04/06/09	FR-051/052	Should

DR-047	Structured data captured for process analysis should have a defined business purpose or identified operational, control, customer-service or measurement use.	Step 23F Data Governance	FR-051	Should
DR-048	The service shall enable authoritative source information to be distinguished from derived, summarised or generated information.	Source-of-truth model, LAC-02	Multiple	Must

DR-049	Where an approved AI capability is used, the service shall maintain an appropriate AI-assistance record linking the use case, authorised source information, output, human review outcome and fallback/correction information required for evaluation and traceability.	ENT-18, AIU-01-06	FR-050	Must — when AI used
---------------	---	-------------------	--------	----------------------------

DR-050	Critical data elements shall have sufficiently defined meaning, validation expectations and permitted values or quality rules to support reliable process execution and analysis.	Data-quality model	Multiple	Must
---------------	---	--------------------	----------	-------------

126.1 AI Assistance Record — Logical Content

- AI Interaction ID
- Case ID
- AI use-case ID
- activity/task type
- authorised source references
- generated timestamp
- generated output/reference
- human reviewer
- review status
- correction/rejection reason where relevant
- material error flag
- material omission flag
- fallback used
- model/service version when later technically applicable

DR-049 does not require AI to be implemented. It specifies the minimum traceability expected **if an approved AI use case is later introduced**.

126.2 AI Review-State Values

Logical values:

- Generated
- Awaiting Human Review
- Accepted
- Accepted with Correction
- Rejected
- Fallback Used
- Flagged for Evaluation

127. Data Quality Framework

Dimension	Application to PayFix
Completeness	Required data is present for the applicable stage
Accuracy	Data reflects the relevant source correctly
Consistency	Equivalent information is not unnecessarily contradictory
Timeliness	Information is available when the process requires it
Validity	Values conform to defined formats/rules
Uniqueness	Duplicate case/event processing is appropriately controlled
Traceability	Origin and lifecycle can be followed
Provenance	Authoritative source is identifiable

Accessibility	Authorised users can obtain information required for their role
Security	Information is protected from inappropriate access

Security and detailed access-control requirements will be formalised in Step 24G rather than being duplicated as Data Requirements.

128. Authoritative Information / Source-of-Truth Matrix

Information	Logical Authoritative Source
Case ID	Case Management
Current case state	Case Management / Workflow
Current queue/owner	Workflow / Ownership record
Original customer statement	Captured customer information
Transaction facts	Authoritative transaction/payment source
Evidence	Original evidence/evidence record
Fraud finding	Fraud Review record
Finance approval	Approval record
Technical finding	Technical Review record
Complaint outcome	Complaints record
Customer communication history	Communication/case record
Final dispute decision	Authoritative Decision Record
Closure	Closure record
AI summary	Derived — not authoritative source
AI recommendation	Derived — not authoritative decision

128.1 Information Classification

Authoritative

Information accepted as the source business record for its purpose.

Derived

Information calculated, summarised or transformed from authoritative data.

Generated

Information created by an AI or comparable generative capability.

Reference

Information linked from another authoritative source.

A derived or generated item can be operationally useful without becoming the authoritative source.

129. ENT-01–ENT-20 → Data Requirement Coverage

Logical Entity	Data Requirements
ENT-01 Dispute Case	DR-001–006
ENT-02 Customer Interaction	DR-010, DR-035
ENT-03 Transaction Reference	DR-008
ENT-04 Evidence Item	DR-011–013
ENT-05 Case Event	DR-039–040, DR-044
ENT-06 Ownership Record	DR-018–019
ENT-07 Waiting Record	DR-022
ENT-08 Handoff Record	DR-020–021, DR-024
ENT-09 Rework Record	DR-023
ENT-10 Specialist Referral	DR-026–028, DR-030
ENT-11 Specialist Finding	DR-029–030
ENT-12 Decision Record	DR-032–033

ENT-13 Approval Record	DR-034
ENT-14 Communication Record	DR-035
ENT-15 Complaint Record	DR-036
ENT-16 Rule Execution Record	DR-041
ENT-17 Automation Execution Record	DR-042
ENT-18 AI Assistance Record	DR-049
ENT-19 Override Record	DR-043
ENT-20 Closure Record	DR-037–038

130. Functional Requirement → Data Requirement Traceability

Functional Area	Main Data Requirements
FR-003 Case creation	DR-001, DR-003
FR-004 Customer/transaction linkage	DR-007–008
FR-005 Customer description	DR-009
FR-007–009 Completeness	DR-014–015
FR-010–011 Customer action	DR-016–017
FR-012–017 Classification/routing	DR-025, DR-041
FR-018–019 Ownership	DR-018–021
FR-020–025 Investigation/evidence	DR-008, DR-011–013, DR-023
FR-027–034 Specialist	DR-026–031
FR-035–038 Decisions/approvals	DR-032–034
FR-039–043 Communication	DR-035
FR-044–046 Complaints	DR-036
FR-047 Closure	DR-037–038

FR-048 Override	DR-043
FR-049 Automation failure	DR-042
FR-050 AI fallback	DR-049
FR-051/052 Monitoring	DR-022-025, DR-039-046

131. Information Flow → Data Requirement Traceability

Flow	Main Data Requirement
IF-TB01 Customer → CS	DR-007, DR-009-010
IF-TB02 CS → Case Management	DR-001-010
IF-TB03 Case → Rules	DR-014-016, DR-041
IF-TB04 Rules → Workflow	DR-041
IF-TB05 Workflow → Operations	DR-018-021
IF-TB06 Transaction source → Ops	DR-008
IF-TB07 Evidence → Ops	DR-011-013
IF-TB08 Ops → Customer	DR-015-017, DR-035
IF-TB09 Customer → Evidence	DR-011-013
IF-TB10 Operations → Specialist	DR-026-028
IF-TB11 Specialist → Case	DR-029-030
IF-TB12 Operations → Finance	DR-026-028, DR-034
IF-TB13 Finance → Case	DR-034
IF-TB14 Case → Communication	DR-032, DR-035
IF-TB15 Communication → Customer	DR-035
IF-TB16 Workflow → Events	DR-039-044
IF-TB17 Events → Analytics	DR-045-046

IF-TB18 Sources → AI	DR-048–049
IF-TB19 AI → Human	DR-049
IF-TB20 Human Review → Record	DR-049

132. KPI → Required Data

KPI	Required Data
SLA breach	DR-003/004 + defined SLA logic
Resolution time	DR-003/004
Handoffs	DR-020
Rework	DR-023
Waiting duration	DR-022
Customer contacts	DR-035
Specialist duration	DR-026/030
Specialist clarification	DR-027/029 + rework data
Closure exceptions	DR-037/043
Routing correction	DR-024
AI acceptance/rejection	DR-049
AI error/omission	DR-049

KPI calculation rules and denominators must be documented separately from raw data fields.

132.1 Denominator Integrity

Metrics must use explicitly defined eligible populations.

For example, the existing SLA breach rate is based on **finalised cases**, rather than classifying pending cases as “No breach.”

Future reporting requirements must preserve similarly explicit denominator definitions.

133. Candidate Controlled Data Domains

- Lifecycle status
- Dispute category
- Priority
- Customer-action state
- Waiting reason
- Handoff reason
- Rework reason
- Specialist type
- Referral reason
- Decision type
- Approval status
- Communication type
- Actor type
- Override type
- AI review status

Controlled domains should be sufficiently specific to support meaningful analysis without creating an excessive or unusable list of values.

133.1 Missing-Value Principle

Where analytically important, the data design should distinguish between:

- not applicable;
- genuinely unknown;
- not yet available;
- not captured due to a data-quality issue.

These states should not be collapsed into a single ambiguous blank value where doing so would materially affect process interpretation or measurement.

134. Future Logical Data Dictionary Structure

Column	Purpose
--------	---------

Data Element ID	Stable identifier
Data Element	Business name
Definition	Meaning
Entity	Parent logical entity
Source	Authoritative source
Data Type Concept	Text/date/status/reference etc.
Required?	Conditionality
Allowed Values	Where controlled
Business Owner	Logical ownership
Used By	Process/function
Validation Rule	Quality expectation
Sensitive?	Later privacy/security assessment
Retention	Later policy requirement
Notes	Assumptions

The full physical data dictionary will be created only when the solution design requires that level of specification.

135. Logical Data Ownership

Data ownership should follow the business meaning of the information rather than being assigned only to the technology storing it.

For example, Technology may operate the platform storing a Fraud finding, while the Fraud business function remains responsible for the business meaning and appropriate use of that finding.

Information	Logical Business Responsibility
Customer-facing case information	Customer Service / Process

Operational case information	Payment Operations / Process
Fraud finding	Fraud
Refund approval	Finance
Technical finding	Technical
Complaint outcome	Complaints
Process-event definitions	Process Owner
AI evaluation information	Data/AI + Process Owner

136. Data Minimisation

Information should be captured, shared and processed because it supports a defined process, user, control, audit or measurement need.

The To-Be design should not collect additional information solely because it might later be useful for analytics or AI.

AI access is particularly subject to task-specific data scope rather than broad access to all case information.

137. AI Data Boundary

Where an AI use case is later approved, only information appropriate to that specific task should be supplied.

AI output must remain linked to the authoritative source information required for human verification and must not silently alter original case facts, evidence or specialist findings.

AI Use Case	Logical Data Needed
AIU-01 Intake	Customer description + approved intake context
AIU-02 Evidence Summary	Selected evidence + provenance

AIU-03 Handoff Summary	Structured case/referral history
AIU-04 Investigation Copilot	Authorised case/evidence/transaction context
AIU-05 Communication Draft	Approved outcome + communication context
AIU-06 Classification Support	Description + approved classification framework

137.1 AI Data Exclusions

AI should not automatically receive:

- unrelated customer cases;
- unrelated transaction data;
- unnecessary personal information;
- unrestricted specialist data;
- unrelated additional-support information;
- data outside the approved AI task.

“More data” is not automatically a valid business requirement for better AI

138. Information Exchange Integrity

For material exchanges:

1. preserve Case ID;
2. preserve source/provenance;
3. validate required inputs;
4. detect material failures;
5. support reconciliation;
6. avoid duplicate customer-impacting actions;
7. retain relevant execution result;
8. maintain authoritative-source distinction.

Examples relate to:

- transaction retrieval;
- evidence retrieval;

- specialist referral;
- approval;
- customer communication;
- process-event capture.

139. Data Failure and Exception Scenarios

ID	Data Scenario	Expected Data Behaviour
DRE-01	Transaction information unavailable	Availability/problem state visible
DRE-02	Evidence missing	Missing evidence not silently treated as present
DRE-03	Evidence retrieval fails	Failure remains visible
DRE-04	Ownership unknown	Explicit exception requiring correction
DRE-05	Duplicate event received	Duplicate processing appropriately controlled
DRE-06	Specialist finding missing	Case not falsely treated as specialist-complete
DRE-07	Approval result missing	Approval remains outstanding
DRE-08	Communication status unavailable	Delivery/record state not invented
DRE-09	AI provenance incomplete	AI output not relied upon where verification requires provenance

DRE-10	Required reason not captured	Data-quality exception visible
--------	------------------------------	--------------------------------

140. Data Requirement Dependencies

Deterministic routing

Depends on:

DR-014 + appropriate classification/routing input + DR-041

Ownership measurement

Depends on:

DR-018–DR-020

Waiting analysis

Depends on:

DR-022

Specialist analysis

Depends on:

DR-026–DR-030

Decision traceability

Depends on:

DR-032–DR-034

Process monitoring

Depends on:

DR-039–DR-046

AI evaluation

Depends on:

DR-048–DR-049

141. Data Requirement Tensions

Range	Design Consideration
More structured data vs staff effort	Capture only useful information
Detailed referral vs data minimisation	Task-relevant context only
Auditability vs usability	Automate event capture where appropriate
Customer status vs restricted information	Separate customer-safe status from internal detail
Evidence reuse vs stale information	Maintain provenance/current state
Measurement detail vs poor reason-code quality	Govern categories
AI context vs privacy	Minimum authorised task scope
Historical traceability vs storage burden	Later retention policy needed

143. Data Requirements Register

Range	Requirement Area	Count
DR-001-006	Case Identity & Lifecycle	6
DR-007-013	Customer, Transaction & Evidence	7
DR-014-017	Completeness & Customer Action	4
DR-018-025	Ownership, Waiting, Handoff & Rework	8
DR-026-031	Specialist Information	6
DR-032-038	Decisions, Approvals, Communications & Closure	7

DR-039-044	Events, Rules, Automation & Overrides	6
DR-045-050	Measurement, Provenance, AI & Data Quality	6
Total		50

143.1 Data Requirement Priority Summary

Priority	Count
Must	43
Should	7
Could	0
Won't	0
Total	50

The Should requirements are:

- DR-013
- DR-017
- DR-024
- DR-031
- DR-045
- DR-046
- DR-047

The high Must count reflects that many Data Requirements are required to support the already-defined core process and control model. This does not mean all data must be implemented in one delivery increment.

143.2 Data Requirement Status

Initially set:

Proposed

Then complete:

- entity coverage;
- FR traceability;
- information-flow traceability;
- quality review;
- source-of-truth review;
- AI boundary review;
- duplication review.

After those checks, change to:

Validated — Portfolio

Validated — Portfolio means internally consistent with the project evidence and design. It does not represent approval by a real Data Owner or organisation.

144. Example Data Acceptance Decomposition

DR-020 — Handoff

Possible future acceptance:

Given an active case is transferred from one operational responsibility to another,
when the transfer is completed,
then the previous responsibility, new responsibility, transfer time and handoff reason can be identified.

DR-022 — Waiting

Given a case enters a material waiting state,
when the dependency later resolves,
then the waiting reason, start and end can be identified.

Given an approved AI summary is generated,
when an authorised reviewer evaluates it,
then the source references and review result remain traceable to the AI interaction.

145. BA Interpretation

The Data Requirements convert the To-Be information model into formal, traceable business-data needs without prematurely defining a physical database architecture.

Particular emphasis is placed on preserving authoritative information, lifecycle history and the reasons behind waiting, handoffs and rework. This addresses an important limitation in the As-Is evidence, which could often show that an event occurred without explaining sufficiently why it occurred.

The requirements also establish structured specialist, decision, approval, communication and event information needed for operational traceability and later KPI evaluation.

AI data is deliberately additive rather than foundational. AI-generated information remains distinguishable from authoritative sources, and the core case-data model remains fully usable without AI.

146. Conclusion

Fifty formal Data and Information Requirements have been defined for PayFix AI 2.0.

The requirements cover case identity and lifecycle, customer and transaction references, evidence provenance, completeness, customer action, ownership, waiting, handoffs, rework, specialist referrals and findings, decisions, approvals, customer communications, complaints, closure, process events, deterministic rules, automation execution, human overrides, performance measurement and conditional AI traceability.

All twenty logical business data entities defined during the To-Be systems analysis now have formal Data Requirement coverage.

The requirements preserve the distinction between authoritative, derived and AI-generated information, while strengthening the process data necessary to measure waiting, handoff, rework and operational friction.

The logical information model remains technology-neutral and does not prescribe a database, API structure, vendor or cloud platform.

The next requirement stage will focus specifically on how users experience and interact with these capabilities through UX, accessibility and communication requirements.

147. UX, Accessibility and Communication Requirements

UX, Accessibility and Communication Requirements define how customers and operational users should be able to understand, interact with and progress through the PayFix AI 2.0 future dispute service.

The requirements are derived from the User Research and UX Discovery outputs, UN-01–UN-19, ASN-01–ASN-10, customer-journey analysis, User Requirements, Functional Requirements and the To-Be service design.

These requirements describe logical experience outcomes rather than prescribing a final user interface, visual design, channel or technology.

The requirements remain portfolio-validated hypotheses. They have not been confirmed through real-user usability testing, accessibility testing or customer interviews.

147.1 UX Requirement Standard

UX requirements should describe an observable user-experience need or service behaviour.

Where practical, requirements should enable later evaluation through usability testing, accessibility review, user walkthroughs, content testing or UAT.

Requirements should avoid assuming that all customers have the same capability, circumstances, channel preference or need for support.

148. Intake and Guidance Requirements

ID	Requirement	Parent / Source	Priority
----	-------------	-----------------	----------

UXR-001	The service shall provide understandable guidance about the information required to begin a dispute.	UR-CUS-001, FR-001/005	Must
UXR-002	The service shall distinguish information required to progress the current stage from information that may only become necessary later.	UR-CUS-001, FR-007	Must
UXR-003	The service shall avoid requiring a customer to understand internal operational terminology in order to initiate a dispute.	UXF, CJ findings	Must

UXR-004	Where a dispute description does not fit a standard category, the customer shall still have a route to submit or obtain assistance rather than being forced to select an inaccurate option.	BR-003/009/012, FR-013/014	Must
UXR-005	Following successful dispute initiation, the customer shall receive understandable confirmation that the dispute has been received.	UR-CUS-002, FR-006	Must
UXR-006	Submission confirmation shall provide appropriate next-step information rather than only a technical acknowledgment or case number.	UR-CUS-002, BR-008	Must

The objective is to make intake sufficiently clear without turning initial submission into an unnecessarily large evidence-gathering exercise.

149. Status and Customer-Action Requirements

ID	Requirement	Parent / Source	Priority
UXR-007	The service shall present customer-facing dispute status in language meaningful to the customer rather than relying solely on internal queue or workflow terminology.	UR-CUS-003, FR-039/041	Must
UXR-008	Where customer action is required, the service shall make that state clearly distinguishable from a state where no customer action is required.	UR-CUS-004, FR-010/040	Must

UXR-009	Where customer action is required, the communication shall explain what action is required and, where appropriate, why it is needed.	UR-CUS-004, FR-011/040	Must
UXR-010	A customer shall not be presented with internal waiting or specialist terminology without sufficient customer-facing explanation of what that means for their journey.	UR-CUS-003/007	Should
UXR-011	Where a case remains open for an extended period or enters specialist processing, appropriate progress information should be available without falsely implying progress where none has occurred.	UR-CUS-003/007, FR-041	Should

149.1 Status Transparency Boundary

Customer-facing status must balance transparency with legitimate Fraud, security, privacy and internal-control restrictions.

“Meaningful status” does not mean exposing unrestricted internal investigation detail.

150. Evidence and Customer-Effort Requirements

ID	Requirement	Parent / Source	Priority
UXR-012	An additional-information or evidence request shall clearly identify what is required from the customer.	UR-CUS-001/004, FR-011	Must
UXR-013	Where appropriate, an evidence request shall explain why the information is needed sufficiently for the customer to understand the request.	UR-CUS-001/004	Must

UXR-014	The customer should not be required to re-enter or resend information already appropriately available to the service unless there is a legitimate reason for requesting it again.	UR-CUS-009, FR-024/025	Should
UXR-015	Where previously supplied information must legitimately be requested again, the service should provide sufficient context to distinguish the request from an accidental duplicate request.	UR-CUS-009, DR-013/025	Should

151. Assisted-Service and Accessibility Requirements

ID	Requirement	Parent / Source	Priority
----	-------------	-----------------	----------

UXR-016	The service shall provide an appropriate assisted route where a customer cannot reasonably complete the standard journey without support.	UR-CUS-005/006, FR-001/002	Must
UXR-017	A customer using an assisted route shall not be prevented from progressing solely because they are unable to use the standard self-service pathway.	BR-009, ASN analysis	Must
UXR-018	The future service shall retain access to appropriate human assistance for cases where automated or self-service interaction is insufficient.	UR-CUS-006, BR-009/012	Must

UXR-019	Support requirements expressed by a customer shall be handled without requiring the service to infer disability, vulnerability or other personal characteristics from unrelated data.	ASN, AIR risks	Must
UXR-020	The service shall support appropriate alternatives or exception handling where a standard interaction method creates a barrier to progressing the dispute.	UR-CUS-005/006, FR-048	Must
UXR-021	Important instructions, actions and outcomes shall not depend solely on visual cues that could be unavailable or unclear to some users.	Accessibility analysis	Must

UXR-022	Accessibility and assisted-service behaviour shall be capable of being validated through appropriate future user and accessibility testing.	UX Discovery evidence gaps	Should
----------------	---	----------------------------	---------------

151.1 Additional-Support Integrity Principle

The portfolio uses “additional support” or “assisted service” as service-design concepts.

It does not infer a specific medical condition, disability or personal characteristic from the synthetic Vulnerable Support segment.

Future service design should respond to an expressed or appropriately identified service need rather than using speculative personal inference.

151.2 Inclusive Journey Principle

- provide a human route where appropriate;
- avoid unnecessary channel lock-in;
- support understandable language;
- make customer action explicit;
- provide recovery from input/error states;
- avoid punishment for AI/automation failure;
- preserve customer progress where possible;
- avoid requiring unnecessary repeated disclosure;
- test exception routes, not only the happy path.

152. Customer Communication Requirements

ID	Requirement	Parent / Source	Priority
----	-------------	-----------------	----------

UXR-023	Customer communications shall clearly identify the dispute or relevant case context to which the communication relates.	BR-008/011, FR-043	Must
UXR-024	Customer communications shall distinguish information, required customer action and final outcome where those concepts differ.	UR-CUS-003/004/008	Must
UXR-025	Final outcome communication shall explain the outcome in language intended to be understandable to the customer.	UR-CUS-008, FR-042	Must

UXR-026	Final outcome communication shall provide appropriate next-step, complaint or escalation information where applicable.	UR-CUS-008, BR-013	Must
UXR-027	Customer communications should use consistent terminology for important lifecycle concepts across the journey.	UXF, communication design	Should
UXR-028	Customer communication content shall not present unverified generated content as authoritative fact or decision information.	BR-011/016, DR-048/049	Must

152.1 Communication Types Requiring Future Content Design

- dispute receipt;
- missing-information request;
- evidence receipt;
- customer-action reminder;

- material progress/status;
- specialist progress where appropriate;
- final dispute outcome;
- refund/action information;
- complaint outcome;
- closure confirmation;

Step 24F defines communication requirements, not final message wording.

152.2 Communication Content Principles

Communications should be accurate, understandable, proportionate to the situation and consistent with the authoritative case state.

Where a customer must act, the communication should prioritise what action is required and what happens next.

Where no action is required, communications should avoid language that unnecessarily implies the customer must respond.

153. Operational User Experience Requirements

ID	Requirement	Parent / Source	Priority
UXR-029	An operational user shall be able to understand the current case status, responsibility state and outstanding customer action without reconstructing the entire event history.	UR-CS-001, UR-OPS-001	Must

UXR-030	A receiving operational or specialist user shall be able to identify the reason for a handoff/referral and the action requested from them.	UR-OPS-003, UR-FRA/FIN/TEC	Must
UXR-031	Relevant evidence and source information shall be presented or accessible in a manner that enables an authorised user to distinguish source evidence from summaries or generated information.	UR-OPS-001, DR-048	Must
UXR-032	Operational users shall be able to identify when a case is waiting and, where appropriate, the dependency or next action associated with the wait.	UR-SVC-001, DR-022	Must

UXR-033	Exceptions, overrides and technology fallbacks shall provide enough context for an authorised user to understand what action is required rather than sending the case to an unexplained generic failure state.	BR-012, FR-048–050	Must
----------------	--	--------------------	-------------

153.1 Operational Usability Principle

Adding more structured information does not automatically improve operational usability.

The future interface should expose the information required for the user's task while avoiding unnecessary cognitive load or forcing staff to search through unrelated case information.

154. Automation and AI Interaction Requirements

ID	Requirement	Parent / Source	Priority
----	-------------	-----------------	----------

UXR-034	Where automation performs a material user-visible activity, the resulting service state shall be sufficiently clear for an authorised user to understand what has occurred or what requires attention.	BR-012/015, FR-049	Must
UXR-035	Where an operational user receives AI-generated assistance, the interface or interaction shall make the AI-assisted nature of the output distinguishable where necessary for appropriate review.	BR-016, DR-049	Must when AI used

UXR-036	Where AI-generated assistance requires human review, the user shall be able to accept, correct, reject or bypass the output as appropriate to the approved use case.	BR-016, GW-18, FR-050	Must when AI used
----------------	--	-----------------------	--------------------------

These requirements do not require AI implementation. They define minimum user-experience expectations if approved AI assistance is later introduced.

154.1 AI Interaction Principle

AI should reduce user effort only where it can do so without obscuring the authoritative information, decision owner or normal non-AI route.

A reviewer should not be forced to accept an AI output simply because the capability is available.

155. UX, Accessibility and Communication Requirements Register

Range	Requirement Area	Count
UXR-001–006	Intake & Guidance	6
UXR-007–011	Status & Customer Action	5
UXR-012–015	Evidence & Customer Effort	4
UXR-016–022	Assisted Service & Accessibility	7

UXR-023-028	Communication & Outcomes	6
UXR-029-033	Operational UX	5
UXR-034-036	Automation / AI Interaction	3
Total		36

155.1 Priority Summary

Priority	Count
Must	28
Should	6
Must when AI used	2
Total	36

The Should requirements are:

UXR-010, 011, 014, 015, 022, 027

Conditional AI requirements:

UXR-035, UXR-036

156. User Requirement → UX Requirement Traceability

User Requirement	Main UXR Coverage
UR-CUS-001	UXR-001-004, 012-013
UR-CUS-002	UXR-005-006
UR-CUS-003	UXR-007, 010-011
UR-CUS-004	UXR-008-009, 012-013, 024
UR-CUS-005	UXR-016-022
UR-CUS-006	UXR-016-020

UR-CUS-007	UXR-010-011
UR-CUS-008	UXR-024-026
UR-CUS-009	UXR-014-015
UR-CS-001	UXR-029, 032
UR-OPS-001	UXR-029, 031
UR-OPS-002	UXR-030
UR-OPS-003	UXR-030-031
UR-FRA-001	UXR-030-031
UR-FIN-001	UXR-030-031
UR-TEC-001	UXR-030-031
UR-COM-001	UXR-029-031
UR-SVC-001	UXR-029, 032-033
UR-SVC-002	UXR-032-033

157. Functional Requirement → UX Requirement Coverage

Functional Area	UX Requirement
FR-001/002 initiation	UXR-001-006, 016-020
FR-006 confirmation	UXR-005-006
FR-007-011 completeness/action	UXR-001/002, 008/009, 012-015
FR-013/014 classification exception	UXR-004
FR-018/019 ownership	UXR-029-030
FR-020-025 investigation/evidence	UXR-012-015, 031
FR-027-034 specialist	UXR-010/011, 030-032
FR-039-043 communication	UXR-007-011, 023-028
FR-044-046 complaints	UXR-025-026

FR-048-050 fallback	UXR-018-020, 033-036
FR-051 monitoring	UXR-032-033

158. Accessibility Discovery → UX Requirement Coverage

Discovery Theme	Requirement Response
Assisted route required	UXR-016
Standard route may be unsuitable	UXR-017
Human support retained	UXR-018
Avoid inferred vulnerability	UXR-019
Flexible exception path	UXR-020
Multiple ways to understand instructions	UXR-021
Test assisted journey	UXR-022
AI must not become only route	UXR-018, 036

159. Customer Journey → UX Requirements

Journey Stage	Main UXR
Issue / initiation	UXR-001-006
Information/evidence	UXR-001-004, 012-015
Routing/investigation	UXR-007-011
Customer waiting	UXR-007-011
Specialist review	UXR-010-011
Additional support	UXR-016-022
Decision/outcome	UXR-023-028
Complaint	UXR-025-028

Closure	UXR-023-027
---------	-------------

160. Customer Communication State Model

Service State	Customer Communication Need
Submitted	Confirmation / next step
Customer action required	Action + reason
Evidence received	Confirmation where appropriate
Investigation active	Meaningful status where appropriate
Specialist activity	Appropriate progress information
Waiting — no customer action	Avoid implying customer must act
Decision reached	Outcome + explanation
Complaint available	Appropriate route
Complaint concluded	Complaint outcome
Closed	Closure information where appropriate

Communication frequency and channel are not yet fixed requirements and should be validated rather than invented.

161. Understandable Content Principle

Customer-facing terminology should be understandable to the intended user and should avoid unnecessary internal operational language.

Important information should communicate:

what has happened → whether the customer must act → what happens next.

The detailed content style and wording remain subject to future content design and testing.

162. Future Accessibility Validation Areas

Area	Validation Question
Intake	Can users understand what information is required?
Errors	Can users recover from incomplete/incorrect input?
Evidence	Are requests understandable?
Status	Is customer action obvious?
Navigation	Can users progress without unnecessary confusion?
Assisted route	Can support be accessed when required?
Communication	Are important instructions understandable?
Outcome	Can users understand the result and next steps?
Automation fallback	Can the journey continue?
AI interaction	Can staff distinguish and review AI output?

163. Potential UX Measures

Measure	Current Evidence
Intake completion/usability	Not measured
Information-completeness rate	Future baseline
Repeat evidence-request rate	Future baseline
Status-enquiry contacts	Future baseline
Customer-action misunderstanding	Future research
Assisted-journey completion	Future research
Customer effort	Future research
Outcome comprehension	Future testing
Operational case reconstruction effort	Future testing
Specialist referral usability	Future walkthrough
AI review usability	Future pilot

164. UX Requirement Tensions

Tension	Relevant Requirements
Simple intake vs adequate information	UXR-001-004
Status transparency vs Fraud confidentiality	UXR-007-011
Reduce repetition vs legitimate evidence re-request	UXR-014/015
Standardisation vs assisted-service flexibility	UXR-016-020
Detailed explanation vs communication overload	UXR-023-027

Rich operational context vs cognitive load	UXR-029-033
AI assistance vs reviewer over-reliance	UXR-035/036
Automation efficiency vs human availability	UXR-018, 034

These tensions should be explored through prototype and usability evidence rather than resolved through assumption alone.

165. UX Requirements Not Supported by Current Evidence

Proposed Requirement	Position
Every customer requires a mobile app	Not established
Every customer prefers chatbot support	Not established
AI chatbot must be the first contact route	Rejected
Customers should never contact a human	Rejected
Every internal status should be shown to customers	Rejected
Fraud investigation detail should always be disclosed	Rejected
Every communication must use one specific channel	Not established
More notifications always improve UX	Not established

Every customer should follow identical steps	Rejected
AI-generated communication can bypass human review in sensitive cases	Rejected

166. Future Prototype Priorities

The first prototype should focus on high-risk experience areas rather than visual polish.

Priority scenarios:

P1 — Dispute initiation and completeness

Test:

- what information is required;
- ability to continue;
- missing information.

P2 — Customer-action state

Test:

- action required;
- reason;
- next step.

P3 — Long-running/specialist status

Test:

- understandable progress;
- no inappropriate disclosure.

P4 — Evidence request

Test:

- clarity;
- repetition;
- receipt.

P5 — Assisted route

Test:

- human support access;
- exception handling.

P6 — Final outcome

Test:

- explanation;
- next steps;
- complaint path.

P7 — Operational case view

Test:

- ownership;
- evidence;
- status;
- handoff.

P8 — AI review interaction

167. UX Questions Requiring Future Evidence

1. Which status descriptions are genuinely meaningful to customers?
2. What level of specialist-progress information is useful without causing confusion?
3. Which evidence-request explanations reduce uncertainty?
4. When do customers prefer or require human assistance?
5. What barriers exist in the standard dispute journey?
6. Which information can be reused without creating trust concerns?
7. How much outcome explanation is understandable and useful?
8. How should customer-action states be communicated?
9. What information do operational users need immediately versus on demand?
10. How should AI-assisted outputs be presented to encourage appropriate review rather than automation bias?
11. Does AI assistance actually reduce user effort once review effort is included?
12. Are assisted-service exceptions easy for staff to identify and manage?

These are genuine research questions, not assumed findings.

168. UX Requirement Quality Review

Check each requirement for:

- traceable user need;
- understandable outcome;
- no fake user preference;
- no premature interface design;
- accessibility considered;
- human assistance preserved;
- restricted information respected;
- testability;
- no dependency on AI;
- appropriate priority;
- no conflict with human-control boundaries;
- explicit future-validation need where uncertain.

169. Example UX Acceptance Decomposition

UXR-008 — Customer Action

Possible later acceptance:

Given an active dispute requires customer evidence,
when the customer views or receives the relevant status communication,
then it is clear that customer action is required.

UXR-016 — Assisted Route

Given the standard route is unsuitable for a customer requiring assistance,
when support is requested,
then an appropriate assisted route is available without requiring abandonment of the dispute.

Given an authorised analyst is shown a generated evidence summary,
when they review the case,
then the underlying source evidence remains identifiable and accessible according to their authorised access.

Given an AI-assisted output requires human review,
when the reviewer considers the output unsuitable,
then they can reject or bypass it and complete the underlying task without AI.

170. BA Interpretation

The UX, Accessibility and Communication Requirements translate the earlier user-needs analysis into explicit future-service expectations without prematurely prescribing screens or channels.

The design focuses particularly on understandable intake, meaningful status, explicit customer action, reason-based evidence requests, reduced unnecessary repetition, accessible and assisted journeys, human support and understandable final outcomes.

Internal-user usability is also addressed through clear ownership, handoff context, source-evidence visibility and understandable exception states.

Automation and AI are treated as supporting interaction mechanisms rather than user needs. Where AI is later used, operational users must remain able to identify, review, correct, reject or bypass the generated assistance as appropriate.

Because no real usability study has yet been performed, the requirements provide a structured basis for future prototype and UX validation rather than claiming that the proposed experience is already proven.

171. Conclusion

Thirty-six UX, Accessibility and Communication Requirements have been defined for PayFix AI 2.0.

The requirements cover intake guidance, customer confirmation, meaningful status, customer-action clarity, evidence requests, reduced unnecessary repetition, assisted journeys, human support, accessible interaction, progress communication, understandable outcomes, operational-user usability and conditional AI interaction.

The requirements remain solution-neutral and do not prescribe a specific application, visual design, channel or AI interface.

Accessibility and assisted-service requirements deliberately avoid inferring personal characteristics and preserve an appropriate human route through the dispute service.

All nineteen User Requirements now have relevant UX consideration, while areas requiring real-user research and usability testing have been explicitly identified.

The next requirements stage will formalise Control, Automation, AI and Non-Functional Requirements.

172. Control, Automation, AI and Non-Functional Requirements

This section defines the control, automation, AI-specific and non-functional requirements necessary to support the PayFix AI 2.0 future payment-dispute service.

The requirements are derived from the mandatory gates, To-Be control points, responsibility model, exception architecture, Functional Requirements, Data Requirements and UX requirements.

The purpose is to preserve business accountability, human decision authority, safe exception handling, auditability, accessibility, security, resilience and reliable service operation while allowing appropriate rules, workflow, conventional automation and optional AI assistance.

Numerical production thresholds, organisation-specific approval limits, legal interpretations, vendors and AI models are intentionally not invented where no evidence basis currently exists.

173. Control Requirements

Control Requirements define authority, oversight, exception, approval, audit and decision boundaries within the future service.

They distinguish activities that technology may execute or support from decisions for which accountable human authority remains required.

173.1 Human Decision Authority

ID	Control Requirement	Source	Priority
CR-001	The service shall maintain identifiable accountable human authority for sensitive customer-impacting dispute decisions where such authority is required.	BR-007, TCP-07	Must
CR-002	A Fraud outcome requiring specialist judgement shall remain attributable to an authorised human Fraud role.	FR-036, TCP-04	Must

CR-003	A required refund approval shall remain attributable to an appropriately authorised human Refund / Finance role.	FR-037, TCP-05/06	Must
CR-004	An authoritative complaint outcome shall remain attributable to an appropriate human Complaints role.	FR-046, TCP-08	Must

These requirements do not prohibit technology from preparing information, creating workflow tasks or supporting human decision-making.

173.2 Rule and Exception Controls

ID	Control Requirement	Source	Priority
CR-005	Material deterministic business rules shall have identifiable business ownership and controlled rule definition/versioning.	Step 23D, DR-041	Must

CR-006	Where a deterministic rule allows authorised override, the override shall occur through a defined exception path rather than uncontrolled bypass.	GW-19, FR-048	Must
CR-007	Progression of a materially incomplete case through an authorised exception shall require an identifiable reason and accountable human action.	GW-03, FR-009	Must
CR-008	Manual correction of classification, priority or routing shall be restricted to an appropriate authorised role and remain traceable where material.	OVR-02-04, FR-014/017	Must

173.3 Approval, Closure and Override Controls

ID	Control Requirement	Source	Priority
CR-009	Where independent approval is required by the business process, the future workflow shall preserve that separation rather than automatically collapsing preparation and approval into one activity.	Step 23D, RDP-08	Must
CR-010	Where an approval is required, the dependent activity shall not be represented as approved until the relevant approval condition has been satisfied.	FR-037/038	Must

CR-011	Closure shall be controlled so that applicable decisions, approvals, required communications and outstanding mandatory actions are completed or appropriately accounted for before finalisation.	FR-047, TCP-10	Must
CR-012	Material human overrides shall retain sufficient information to identify the original result, revised result, authorised actor, reason and time where appropriate.	DR-043	Must

173.4 Conventional Automation Controls

Automation is not authorised to invent new business policy because a workflow configuration allows an action technically.

173.5 Service, Access and Control-Governance Requirements

ID	Control Requirement	Source	Priority
----	---------------------	--------	----------

CR-016	An appropriate assisted or human-supported route shall remain available where the standard or automated route is unsuitable for the customer's needs.	BR-009, UXR-016-020	Must
CR-017	Access to customer, case, specialist and decision information shall be limited according to an appropriate authorised role or business need.	Step 23F, LAC-08	Must
CR-018	Material control exceptions, overrides, automation failures and AI-related control events should be capable of periodic review to identify recurring control or process issues.	TB-14, BR-017	Should

174. Control Requirements Register

Range	Area	Count
CR-001-004	Human Decision Authority	4
CR-005-008	Rules & Exceptions	4
CR-009-012	Approval / Closure / Override	4
CR-013-015	Automation Controls	3
CR-016-018	Service / Access / Governance	3
Total		18

175. Non-Functional Requirements

Non-Functional Requirements define the quality attributes and operating characteristics required of the future PayFix AI 2.0 service.

Where exact measurable thresholds are not yet evidenced, the requirement identifies the quality need and requires an appropriate measurable threshold to be established before formal testing or implementation.

175.1 Security and Access

ID	Non-Functional Requirement	Priority
NFR-001	The service shall protect dispute information from unauthorised access.	Must

NFR-002	Access privileges shall follow an appropriate least-privilege principle based on the user's authorised responsibilities.	Must
NFR-003	Sensitive information shall be appropriately protected during storage and exchange using security controls suitable for the eventual architecture.	Must
NFR-004	Privileged or sensitive case actions shall require appropriate authentication and authorisation controls.	Must

175.2 Privacy and Data Minimisation

ID	Non-Functional Requirement	Priority
NFR-005	The service shall support data minimisation so that information captured, shared or processed has an identified business, user, control, audit or measurement purpose.	Must

NFR-006	Information made available to a user, specialist, automation or AI capability shall be limited to an appropriate authorised scope for the relevant task.	Must
----------------	--	-------------

Detailed legal/privacy policy interpretation remains outside the evidence currently available in this synthetic portfolio project.

175.3 Availability, Recovery and Continuity

ID	Non-Functional Requirement	Priority
NFR-007	The core dispute service shall support an availability level appropriate to the agreed operating need, with the measurable target established before formal service acceptance.	Must
NFR-008	Material service or integration failure shall support recovery or a defined contingency sufficient to prevent silent loss of active case work.	Must

NFR-009	Failure of an optional AI capability shall not prevent the core dispute service from operating.	Must
NFR-010	Persistent material case data shall be recoverable to an agreed level appropriate to the eventual service design.	Must

175.4 Reliability and Processing Integrity

ID	Non-Functional Requirement	Priority
NFR-011	Material workflow and automation actions shall operate consistently with the defined case state and business rules.	Must
NFR-012	Where retries or repeated messages can occur, the design shall appropriately control duplicate processing that could create inconsistent or repeated business action.	Must

NFR-013	Material lifecycle, ownership, decision and approval information shall remain internally consistent enough to support reliable process execution.	Must
----------------	---	-------------

175.5 Performance

ID	Non-Functional Requirement	Priority
NFR-014	Critical user and workflow interactions shall meet measurable performance expectations appropriate to the operational task, with thresholds defined before formal performance testing.	Should

No arbitrary response-time figure is assigned at this stage.

175.6 Auditability and Observability

ID	Non-Functional Requirement	Priority
----	----------------------------	----------

NFR-015	Material audit and process-event information shall be sufficiently reliable to support authorised reconstruction of significant case activity.	Must
NFR-016	The service should provide sufficient operational observability to identify material workflow, integration, automation and relevant AI failures.	Should

175.7 Maintainability and Change

ID	Non-Functional Requirement	Priority
NFR-017	Business rules, controlled reason values and workflow configuration should be maintainable through controlled change without unnecessary redesign of unrelated service capability.	Should
NFR-018	Material configuration changes should support versioning, testing and controlled release before affecting the live service.	Should

175.8 Interoperability and Capacity

ID	Non-Functional Requirement	Priority
NFR-019	Information exchanges between logical capabilities should use sufficiently defined and consistent interfaces/contracts to preserve Case ID, provenance and processing integrity.	Should
NFR-020	The service should be capable of supporting an agreed operational workload and reasonable future growth without requiring unsupported assumptions about current transaction volume.	Should

176. Non-Functional Requirements Register

Area	IDs
Security / Access	NFR-001-004
Privacy / Minimisation	NFR-005-006
Availability / Recovery	NFR-007-010
Reliability / Integrity	NFR-011-013
Performance	NFR-014
Audit / Observability	NFR-015-016

Maintainability	NFR-017-018
Interoperability / Capacity	NFR-019-020

Priority	Count
Must	14
Should	6
Total	20

177. AI-Specific Requirements

AI-Specific Requirements apply only if one or more AI use cases are approved to progress beyond evaluation.

They do not make AI mandatory for the service.

These requirements establish the minimum boundaries necessary for AI assistance to remain subordinate to authoritative source information, accountable human judgement and the normal non-AI process.

177.1 AI Scope and Service Independence

ID	AI Requirement	Priority
AIR-001	AI capability shall only be used for an explicitly approved and bounded AI use case.	Must when AI used
AIR-002	No approved AI capability shall be required for completion of the core end-to-end dispute lifecycle.	Must

AIR-003	AI shall receive only the authorised information required for the relevant approved use case.	Must when AI used
AIR-004	AI-generated information shall remain distinguishable from authoritative customer, evidence, specialist and decision information.	Must when AI used

177.2 Source Traceability and Human Review

ID	AI Requirement	Priority
AIR-005	Where AI output depends on case or evidence information, the relevant authorised source information shall remain identifiable for human verification.	Must when AI used
AIR-006	AI output requiring review shall not become authoritative solely because it has been generated.	Must when AI used
AIR-007	An authorised human reviewer shall be able to accept, correct, reject or bypass AI-assisted output where the approved use case requires human review.	Must when AI used

AIR-008	Rejection or unavailability of AI output shall return the activity to an appropriate normal human or non-AI process without penalising the customer.	Must when AI used
----------------	--	--------------------------

177.3 AI Decision Authority Boundaries

ID	AI Requirement	Priority
AIR-009	AI shall not independently determine the authoritative Fraud outcome.	Must
AIR-010	AI shall not independently approve a refund where authorised human approval is required.	Must
AIR-011	AI shall not independently determine the authoritative complaint outcome.	Must
AIR-012	AI shall not independently make a sensitive final dispute decision where accountable human judgement is required.	Must

177.4 AI Traceability, Communication and Evaluation

ID	AI Requirement	Priority
AIR-013	Material AI interactions shall retain sufficient traceability to identify the AI use case, relevant source context, generated output and human review outcome where applicable.	Must when AI used
AIR-014	AI interactions should support identification of the relevant model/service version or configuration where later required for evaluation, investigation or change control.	Should when AI used
AIR-015	AI-generated customer communication used in sensitive or bespoke circumstances shall remain subject to the appropriate human-review requirement before external use.	Must when AI used

AIR-016	An AI use case shall support evaluation and a defined continue, modify or stop decision based on evidence of value, error, omission, human-review burden, control impact and operational risk.	Must when AI used
----------------	--	--------------------------

178. AI Use Case → Requirement Mapping

Use Case	Relevant AIR
AIU-01 Intake Assistance	AIR-001-008, 013-016
AIU-02 Evidence Summary	AIR-001-008, 013-016
AIU-03 Handoff Summary	AIR-001-008, 013-016
AIU-04 Investigation Copilot	AIR-001-008, 012-016
AIU-05 Communication Drafting	AIR-001-008, 013-016
AIU-06 Classification Support	AIR-001-008, 013-016

178.1 AI Human-Review Model

AI Use Case	Review Model
AIU-01 Intake	Draft/recommendation verified by customer/user
AIU-02 Evidence Summary	AI summarises; human verifies against source
AIU-03 Handoff Summary	AI drafts; operational user verifies
AIU-04 Investigation Copilot	AI assists retrieval/analysis; analyst decides

AIU-05 Communication	AI drafts; authorised human reviews where required
AIU-06 Classification	AI recommends; authorised user accepts/rejects/corrects

AI does not become a RACI-accountable participant.

179. AI Evaluation Requirements

- acceptance rate;
- correction rate;
- rejection rate;
- factual errors;
- material omissions;
- human-review effort;
- fallback usage;
- operational time/effort;
- customer-impacting incidents;
- privacy/security issues;
- service availability;
- user trust/appropriate reliance.

AI generation speed alone is not sufficient evidence of business value.

179.1 AI Continue / Modify / Stop Principle

An AI pilot should not automatically progress to broader use because the technology functions technically.

Evidence should support a decision to:

Continue — demonstrated value and acceptable controls;

Modify — potential value exists but material deficiencies require change;

Stop — value is insufficient or risk/control issues make the use case unsuitable.

Numeric thresholds should be agreed before testing rather than invented retrospectively after observing results.

180. Decision and Execution Hierarchy

When selecting how an activity should be performed, PayFix AI 2.0 follows this logical sequence:

1. Simplify the process

↓

2. Structure the required information

↓

3. Use deterministic rules where the logic is stable

↓

4. Use workflow to manage state and ownership

↓

5. Use conventional automation for repeatable execution

↓

6. Consider AI only where it provides additional value

↓

7. Retain human judgement where ambiguity, sensitivity or control requires it

181. To-Be Control Point → Control Requirement Traceability

Control Point	Requirement
TCP-01 Mandatory information	CR-005-007
TCP-02 Classification/routing	CR-005/006/008
TCP-03 Fraud referral	CR-005, CR-017
TCP-04 Fraud outcome	CR-002
TCP-05 Refund authority requirement	CR-005/010
TCP-06 Refund approval	CR-003/009/010
TCP-07 Sensitive outcome	CR-001

TCP-08 Complaint decision	CR-004
TCP-09 Human override	CR-006/012
TCP-10 Closure validation	CR-011
TCP-11 AI output acceptance	AIR-006–008
TCP-12 Access/privacy	CR-017, NFR-001–006

182. Gateway → Control / AI Requirement Traceability

Gateway	Relevant Requirement
GW-03 Incomplete exception	CR-007
GW-04/05 Classification	CR-005/006/008
GW-06 Routing	CR-005/006/008
GW-09 Fraud	CR-002
GW-11 Refund approval	CR-003/009/010
GW-13 Controls complete	CR-010/011
GW-15 Closure	CR-011
GW-16 Automation failure	CR-014, NFR-008
GW-17 AI available/applicable	AIR-001–004
GW-18 AI acceptable	AIR-005–008
GW-19 Override	CR-006/008/012
GW-20 Waiting	NFR-015/016 + DR-022

183. Future Non-Functional Verification

Quality Area	Future Verification
Security	Security assessment / access tests
Authorisation	Role/access test

Data minimisation	Data/design review
Availability	Service test against agreed target
Recovery	Failure/recovery test
AI independence	Disable AI and execute core journey
Reliability	Workflow/integration tests
Duplicate control	Retry/duplicate scenario
Performance	Performance testing against agreed thresholds
Auditability	Event/history reconstruction
Observability	Failure-alert/monitoring test
Maintainability	Configuration-change test
Interoperability	Contract/integration tests
Capacity	Load/capacity testing against agreed demand

184. Requirement Tensions

Tension	Requirements
Fast handling vs required human approval	CR-001-004 / performance needs
Least privilege vs operational context	CR-017 / NFR-001/002
Detailed audit vs data minimisation	NFR-005 / NFR-015
Automation efficiency vs recoverability	CR-013-015 / NFR-008
Rules consistency vs exception flexibility	CR-005-008

AI usefulness vs minimum data scope	AIR-003 / AIR-005
AI speed vs human-review effort	AIR-007 / AIR-016
Customer transparency vs Fraud restriction	UXR status requirements / CR-002/017
Maintainability vs control	NFR-017/018 / CR-005

These tensions require balanced design and testing rather than treating either side as automatically dominant.

185. Requirements Not Supported by Current Evidence

Proposed Requirement	Position
AI may autonomously approve refunds	Rejected
AI may autonomously decide Fraud	Rejected
AI may autonomously decide complaints	Rejected
AI must be available for the core service	Rejected
All automation may bypass human exception handling	Rejected
Every user should access every case field	Rejected
A specific encryption technology must be used	Not established
Service must achieve 99.999% availability	Not established
Every screen must respond in under an invented number of milliseconds	Not established

AI accuracy must be an arbitrary percentage	Not established
AI model/vendor must be OpenAI or another named supplier	Not supported
Specific refund approval thresholds	Not established

186. Step 24G Requirements Summary

Requirement Type	Count
Control Requirements	18
Non-Functional Requirements	20
AI-Specific Requirements	16
Total	54

AIR requirements are conditional where AI capability is introduced. They should not be interpreted as scope commitment to implement all AI use cases.

187. Example Control / NFR / AI Acceptance Decomposition

CR-003 — Refund Approval

Given a dispute requires authorised refund approval,
when the approval activity is completed,
then the approval can be attributed to the appropriate authorised human role before the dependent action is treated as approved.

CR-014 — Automation Failure

Given a required automated workflow activity fails,
when the failure occurs,
then the case remains visible and an appropriate recovery or manual fallback path can be initiated.

Given all optional AI capability is unavailable,
when a valid dispute is processed,
then the core end-to-end dispute journey can still be completed.

Given an AI-assisted output requires human review,
when the reviewer determines that the output is unsuitable,
then they can correct, reject or bypass the output without preventing completion of the underlying business activity.

188. Step 24G Requirement Quality Review

- preserve human accountability;
- distinguish rules, automation and AI;
- permit controlled exceptions;
- preserve required approvals;
- maintain safe fallback;
- protect source evidence;
- protect assisted-service access;
- support traceability;
- avoid invented regulatory claims;
- avoid invented SLA figures;
- avoid invented AI accuracy levels;
- remain technology/vendor neutral;
- are testable once measurable thresholds are agreed;
- do not make AI mandatory;
- do not contradict FR/DR/UXR requirements.

189. BA Interpretation

The Control, Non-Functional and AI-Specific Requirements establish the boundaries within which the future PayFix AI 2.0 service can safely use workflow, rules, automation and optional AI.

Human business authority remains explicit for Fraud outcomes, required refund approvals, sensitive dispute decisions, complaints and authorised exceptions.

Conventional automation is permitted for sufficiently defined repeatable activities but must provide visible failure and recovery behaviour.

Non-functional requirements establish logical expectations for security, privacy, availability, recovery, reliability, auditability, observability, maintainability and interoperability without inventing unsupported production thresholds.

AI requirements preserve source traceability, human review, rejection/correction, service fallback and decision-authority boundaries. AI therefore remains an optional supporting capability rather than the accountable operating model.

190. Conclusion

Eighteen Control Requirements, twenty Non-Functional Requirements and sixteen AI-Specific Requirements have been defined for PayFix AI 2.0.

The requirements formalise human decision authority, rule governance, controlled exceptions, approvals, closure integrity, automation safeguards, assisted-service protection, security, privacy, recovery, reliability, auditability, observability and maintainability.

Conditional AI requirements establish bounded use cases, authorised data scope, source traceability, human review, correction/rejection, non-AI fallback, evaluation and explicit restrictions on autonomous Fraud, refund, complaint and sensitive dispute decisions.

No unsupported production availability target, response-time target, AI accuracy threshold, monetary approval limit, vendor, AI model or infrastructure technology has been invented.

The detailed requirements catalogue is now sufficiently complete to proceed to requirements prioritisation, end-to-end traceability, Jira decomposition and portfolio evidence consolidation.

191. Requirements Consolidation and Baseline

The PayFix AI 2.0 requirements catalogue consolidates the business, user, functional, data, UX/accessibility, control, non-functional and conditional AI requirements derived from Discovery and the approved To-Be process design.

This section performs the final requirements-stage quality, prioritisation, traceability and coverage review before the requirements are used for detailed Jira backlog decomposition, acceptance criteria and testing.

The requirements baseline represents a portfolio-level working baseline. It does not claim formal approval from a real organisation, customer group or business stakeholder.

191.1 Requirements Catalogue Summary

Requirement Type	IDs	Count
Business Requirements	BR-001–BR-017	17
User Requirements	UR-*	19

Functional Requirements	FR-001–FR-052	52
Data Requirements	DR-001–DR-050	50
UX / Accessibility Requirements	UXR-001–UXR-036	36
Control Requirements	CR-001–CR-018	18
Non-Functional Requirements	NFR-001–NFR-020	20
AI-Specific Requirements	AIR-001–AIR-016	16
Total		228

The requirement count is used for catalogue control only. A larger number of requirements does not by itself indicate better analysis, and the 228 requirements must not be converted into 228 individual Jira tickets.

192. Requirements Priority Overview

Priority Classification	Count
Unconditional Must	182
Should	33
Must when AI is used	12
Should when AI is used	1
Total	228

Requirement priority and delivery sequence are separate concepts.

A requirement may be a Must for the eventual viable service while being delivered in a later increment because another capability must be established first.

Conditional AI requirements only become delivery obligations if the corresponding AI use case is approved to proceed.

192.1 Priority vs Delivery Sequence

MoSCoW expresses requirement importance, but it does not determine implementation order by itself.

PayFix AI 2.0 will therefore combine requirement priority with dependencies, control criticality, user need, technical feasibility and the phased solution strategy established during the options appraisal.

193. Requirements Delivery Horizons

Horizon	Purpose
H1 — Core Process Foundation	Process, structured data, workflow, rules, ownership, controls and viable human service
H2 — Selective Automation	Automate stable administrative activities after process/rules are established
H3 — Controlled AI Pilot	Evaluate AIU-02 and AIU-03 against non-AI alternatives
H4 — Conditional AI Expansion	Consider other AI use cases only if evidence justifies them

193.1 H1 — Core Process Foundation

H1 includes the requirements necessary to establish:

- dispute initiation;
- assisted/human initiation;
- case creation;
- information completeness;
- classification;
- routing;
- ownership;
- investigation;
- evidence;
- customer action;

- specialist referral;
- Fraud Review;
- Finance approval;
- Technical Review;
- human dispute decision;
- communication;
- complaints;
- closure;
- core data;
- auditability;
- accessibility;
- safe exceptions;
- core security/privacy;
- reporting-event foundation.

H1 broadly corresponds to the **OPT-02 process/rules/workflow foundation** selected during the options appraisal.

193.2 H2 — Selective Conventional Automation

Candidate automation includes:

- case/reference creation;
- completeness-rule execution;
- standard classification;
- priority application;
- deterministic routing;
- work-item creation;
- standard lifecycle notifications;
- evidence-receipt confirmation;
- SLA/process alerts;
- referral packaging;
- closure-condition checking;
- process-event capture.

Automation should only progress where the underlying process, data, rule and fallback requirements are sufficiently stable.

193.3 H3 — Controlled AI Pilot

Initial candidates remain:

AIU-02 — Evidence Summarisation

and

AIU-03 — Case / Handoff Summarisation

Required supporting requirement areas include:

- DR-048-049;
- UXR-031, UXR-035-036;
- relevant CR/NFR requirements;
- AIR-001-AIR-008;
- AIR-013-AIR-016;
- non-AI fallback;
- human review;
- evaluation evidence.

H3 is an evaluation horizon, not a production-AI commitment.

193.4 H4 — Conditional AI Expansion

Potential later candidates:

- AIU-01 Intake Assistance
- AIU-04 Investigation Copilot
- AIU-05 Communication Drafting
- AIU-06 Exception Classification Support

These capabilities remain conditional. They should not enter implementation merely because earlier AI pilots are technically successful.

194. Core-Service Requirement Boundary

The requirements baseline confirms that PayFix AI 2.0 can satisfy the core payment-dispute lifecycle without AI.

Business value from AI must therefore be demonstrated as **incremental value over the process, data, rules, workflow and conventional-automation foundation**, rather than being assumed as a prerequisite for service operation.

195. Requirements Coverage Review

Evidence / Design Area	Requirement Coverage
Business problems	BR
UN-01-UN-19	UR
TB-01-TB-14	FR
GW-01-GW-20	FR / CR
ENT-01-ENT-20	DR
Customer Journey	UR / UXR
Accessibility	UXR / CR / NFR
TCP-01-TCP-12	CR / AIR
Technology failure	FR / CR / NFR
Process measurement	DR / FR / NFR
Human authority	BR / FR / CR / AIR
Automation	FR / CR / NFR
AI assistance	AIR / DR / UXR / NFR
Closure	BR / FR / DR / CR
Auditability	DR / CR / NFR

No major Discovery or To-Be design area is intentionally left without a corresponding requirement layer.

195.1 Gap Analysis

Check	Result
Do all 17 BRs have downstream solution coverage?	Yes
Do all 19 URs have functional coverage?	Yes
Do all 19 URs have relevant UX consideration?	Yes
Are TB-01–TB-14 represented?	Yes
Are GW-01–GW-20 represented?	Yes
Are ENT-01–ENT-20 represented?	Yes
Are TCP-01–TCP-12 represented?	Yes
Is human fallback represented?	Yes
Is AI-independent operation represented?	Yes
Are future testing links still required?	Yes — later stage
Is real stakeholder validation still required in a live project?	Yes

196. Duplicate and Overlap Review

Overlap between requirement types is expected because Business, User, Functional, Data, UX, Control and Non-Functional requirements describe different aspects of the same business capability.

Requirements should only be merged where they express the same requirement at the same level of abstraction.

Related Requirements	Why They Are Not Duplicates
BR-004 + FR-018 + DR-018	Business capability → behaviour → data
BR-008 + UR-CUS-003 + UXR-007	Business outcome → user need → experience requirement
BR-007 + FR-036 + CR-002	Business authority → function → control
BR-016 + FR-050 + AIR-008	AI independence → fallback behaviour → AI guardrail

197. Requirements Conflict Review

Conflict Area	Resolution Position
Simple intake vs completeness	Stage-appropriate completeness
Standardisation vs exceptions	Deterministic rules + authorised exception
Speed vs Fraud control	Improve surrounding process, retain Fraud judgement
Automation vs human support	Automation cannot remove assisted route
Transparency vs restricted information	Meaningful customer-safe status
Evidence reuse vs changing evidence	Reuse where suitable; legitimate re-request allowed
Auditability vs data minimisation	Capture only traceability with defined purpose
AI usefulness vs privacy	Task-specific authorised context
AI efficiency vs review effort	Measure both
Closure automation vs exceptional cases	Closure rules + human exception

No unresolved requirement conflict currently requires removal of a core requirement. Several areas remain future validation and design questions.

198. Consolidated Requirements Traceability Matrix

Source	BR	UR	FR	DR	UXR	CR/N FR/A IR	Process	Jira	AC/Test
UN-04	BR-005/ 008	UR-CUS-004	FR-010/011/040	DR-016/017	UXR-008/009	Later supporting NFR	TB-07/11	Delivery package	Step 25+

Source	BR	UR	FR	DR	UXR	CR/N FR/A IR	Process	Jira	AC/Test
UN-07 / PP-02	BR-004	UR-CS-001	FR-018/019	DR-018-021	UXR-029/030	NFR-013/015	TB-05	Delivery package	Step 25+

Source	BR	UR	FR	DR	UXR	CR/N FR/A IR	Process	Jira	AC/Test
UN-10 / GATE-01	BR-006/ 007	UR-FRA-001	FR-028/030/034/036	DR-026-032	UXR-030/031	CR-002, AIR-009	TB-08/09F	Specialist package	Step 25+

Sour ce	BR	UR	FR	DR	UXR	CR/N FR/A IR	Proc ess	Jira	AC/T est
UN- 11 / GATE -02	BR- 006/ 007	UR- FIN- 001	FR- 032/ 034/ 037/ 038	DR- 026- 028/ 034	UXR- 030/ 031	CR- 003/ 009/ 010, AIR- 010	TB- 09R/1 0	Speci alist pack age	Step 25+

Sour ce	BR	UR	FR	DR	UXR	CR/N FR/A IR	Proc ess	Jira	AC/T est
UN- 05/0 6 / ASN	BR- 009	UR- CUS- 005/ 006	FR- 001/0 02/0 48- 050	DR- 010	UXR- 016- 022	CR- 016	TB- 01 / exce ption s	Cust omer journ ey pack age	Step 25+

Sour ce	BR	UR	FR	DR	UXR	CR/N FR/A IR	Proc ess	Jira	AC/T est
UN- 14/19	BR- 010/0 11/01 7	UR- SVC- 001/0 02	FR- 051/0 52	DR- 018- 025/ 039- 046	UXR- 032/ 033	NFR- 015/0 16	TB- 14	Data/ repor ting pack age	Step 25+

Sour ce	BR	UR	FR	DR	UXR	CR/N FR/A IR	Proc ess	Jira	AC/T est
------------	----	----	----	----	-----	--------------------	-------------	------	-------------

AIU-02	BR-016	UR-OPS-001	FR-026/050	DR-048/049	UXR-031/035/036	AIR-001-008/013-016, NFR-009	TB-06	AI pilot package	Pilot tests
--------	--------	------------	------------	------------	-----------------	------------------------------	-------	------------------	-------------

The detailed requirement registers remain the authoritative requirement catalogue. The consolidated RTM demonstrates the relationships between requirement layers without duplicating every requirement statement.

Jira and acceptance/test references will be populated progressively during backlog refinement and testing rather than fabricated before those artefacts exist.

199. Requirements Delivery Packages

Package	Scope
DP-01	Dispute Intake and Case Creation
DP-02	Completeness, Classification, Routing and Ownership
DP-03	Investigation and Evidence Management
DP-04	Specialist Referral and Review
DP-05	Decisions, Approvals and Closure
DP-06	Customer Status, Communication and Assisted Service
DP-07	Complaints and Escalations
DP-08	Process Data, Audit, Reporting and Monitoring
DP-09	Rules, Automation, Exceptions and Resilience
DP-10	AI Pilot — Evidence Summarisation

DP-11	AI Pilot — Handoff Summarisation
--------------	----------------------------------

199.1 Package Delivery Sequence

Package	Horizon
DP-01	H1
DP-02	H1
DP-03	H1
DP-04	H1
DP-05	H1
DP-06	H1
DP-07	H1
DP-08	H1 → continuously improved
DP-09	H1 controls + H2 automation
DP-10	H3
DP-11	H3

200. High-Level Requirement Dependencies

DP-01 Intake



DP-02 Classification / Routing / Ownership



DP-03 Investigation / Evidence



DP-04 Specialist Processing



DP-05 Decision / Approval / Closure



DP-06 Customer Communication

Cross-cutting:

DP-07 Complaints

DP-08 Data / Audit / Reporting

DP-09 Rules / Automation / Resilience

Then:

H1/H2 foundation



DP-10 AI Evidence Summary Pilot

DP-11 AI Handoff Summary Pilot

AI pilot packages depend on the process/data foundation; the foundation does not depend on the AI pilot packages.

For each core Epic use a short structure:

Purpose: Deliver the PayFix AI 2.0 capability represented by this requirements package.

Requirement Basis: Business, User, Functional, Data, UX, Control and Non-Functional Requirements defined in the portfolio requirements baseline.

Delivery Approach: Incremental Agile delivery with requirement-to-story and acceptance traceability.

AI Dependency: None unless explicitly stated.

For DP-10 and DP-11 use:

Status: Controlled pilot candidate only.

Dependency: Core process/data foundation must exist first.

Human Review: Required.

Fallback: Normal non-AI process retained.

Go / Modify / Stop: Subject to evaluation evidence.

201. Requirement Package Definition of Ready

A package is ready for detailed story refinement when:

- business purpose is understood;

- relevant BR identified;
- relevant UR identified;
- primary FRs identified;
- necessary DRs identified;
- UX/accessibility implications identified;
- control requirements identified;
- NFR implications understood;
- dependencies identified;
- outstanding assumptions visible;
- major conflicts resolved;
- user/story actor can be identified;
- acceptance can realistically be defined.

For AI packages additionally:

- AI use case approved for evaluation;
- non-AI baseline exists;
- source information defined;
- human reviewer defined;
- fallback defined;
- evaluation approach defined.

Status: Requirements – In Analysis

to:

Requirements Baseline — Portfolio

Keep:

Version: 1.0

Add:

Baseline date: [enter the date you complete Step 24H]

And add:

Approval status: Portfolio baseline only — real stakeholder approval not claimed.

202. Requirements Baseline Statement

Version 1.0 of the PayFix AI 2.0 Business, User and Solution Requirements has completed the portfolio-level quality, coverage, traceability, prioritisation and consistency review.

The baseline contains 228 requirements across eight requirement types.

The baseline is suitable to support backlog decomposition, user-story creation, acceptance criteria and future testing within the portfolio project.

Requirements may still change where future prototype, usability, control, technical or AI evaluation evidence demonstrates a justified need for change. Any material changes should be recorded through the defined requirement change-control process.

This baseline does not represent formal approval by a real PayFix organisation or external stakeholder.

204. Conclusion

The PayFix AI 2.0 Business, User and Solution Requirements stage is complete for the current portfolio baseline.

A total of 228 requirements have been defined across seventeen Business Requirements, nineteen User Requirements, fifty-two Functional Requirements, fifty Data Requirements, thirty-six UX/Accessibility Requirements, eighteen Control Requirements, twenty Non-Functional Requirements and sixteen AI-Specific Requirements.

Requirements have been checked for source traceability, business and user coverage, To-Be process coverage, data-entity coverage, control alignment, priority, dependencies, overlap, conflict, technology neutrality and testability.

Human authority remains explicit for Fraud outcomes, required refund approvals, sensitive dispute decisions, complaint outcomes and authorised exceptions. Assisted and human-supported customer routes are retained.

The core dispute service remains independent of AI. AI requirements apply only to approved use cases and preserve source traceability, human review, rejection/correction, safe fallback and evidence-based continue/modify/stop decisions.

Eleven delivery packages now provide the bridge between the requirements baseline and Agile delivery.

The requirements baseline is therefore ready to progress into detailed Jira User Stories, acceptance criteria, backlog refinement and subsequent testing.